

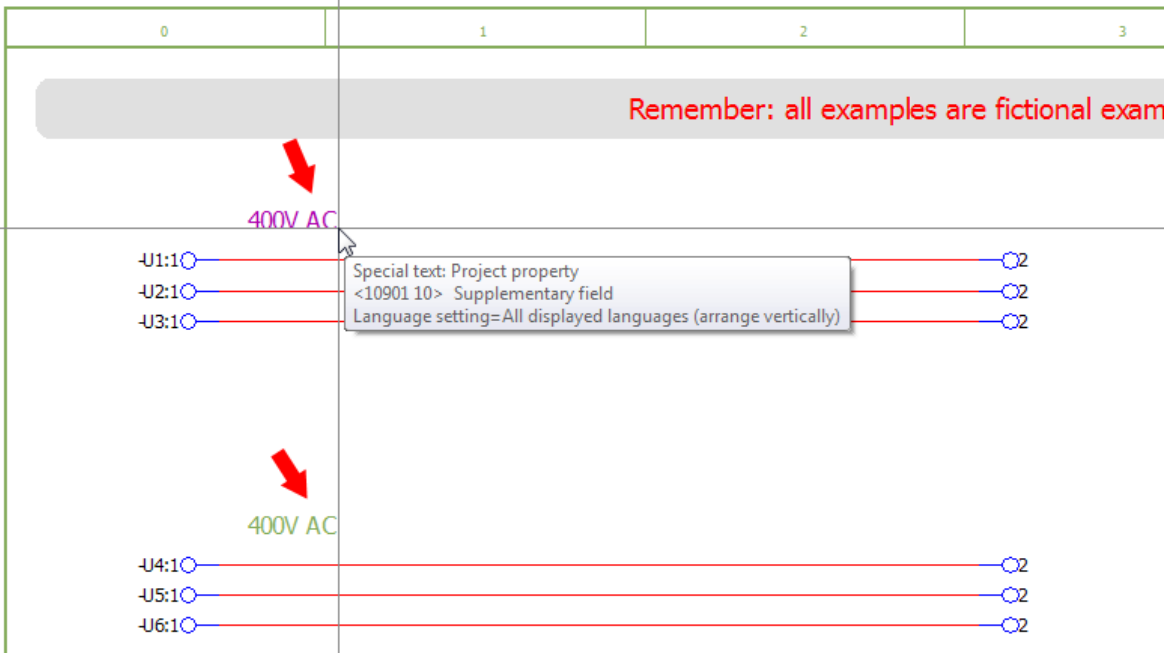


Fig. 4.125 Placed project property

If the voltage later changes or additional information is needed, the input in the project properties can simply be changed, and everywhere in the project where this project property is displayed will be automatically updated. This can save a lot of time in manually changing this information.

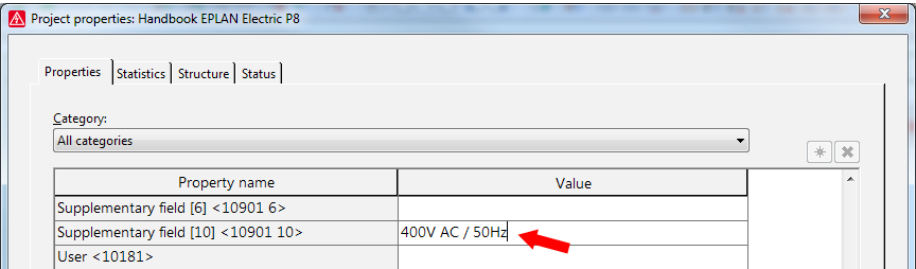


Fig. 4.126 Changed input in the project property

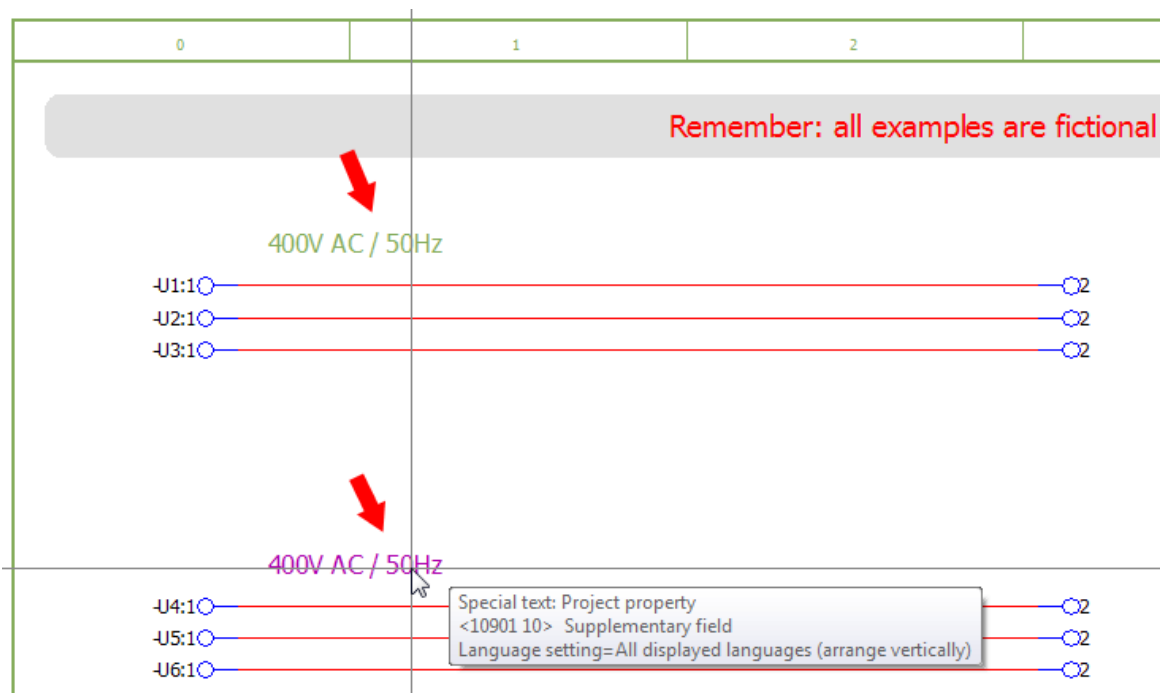


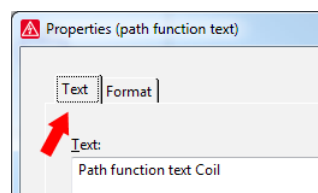
Fig. 4.127 Changed display in the project

4.7.4 Properties – Text dialog

To insert and format texts according to requirements, you have to call up the text's **PROPERTIES – TEXT** dialog. You select text and press **ENTER** key or a left click to open the dialog. The dialogs for normal text and path function text are the same. Only the window title bar of the text input field has a different text type name.

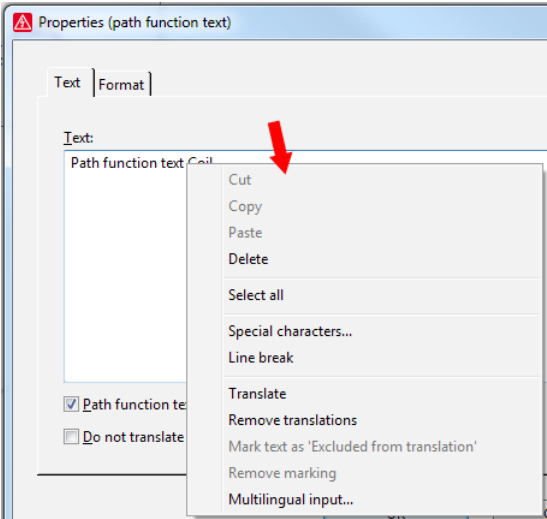
4.7.4.1 Text tab

This *Text* tab contains the *Input field* for the text. You enter text in the *Text* field. You can translate the text at this point, or remove the translation.

Fig. 4.128
Text input field

You call up the popup menu using the right mouse button. In addition to the usual Windows text functions such as COPY, PASTE, DELETE, and CUT, the popup menu also contains other functions such as SPECIAL CHARACTERS or TRANSLATE, REMOVE TRANSLATION, REMOVE MARKING and MULTILINGUAL INPUT.

Fig. 4.129
Popup menu



4.7.4.2 Format tab

The *Format* tab contains all functions needed to format the text (*Font size* or *Font*) or, for example, to assign it a different layer. The formatting options and settings can be divided roughly into the three nodes; *Format*, *Border* and *Value/unit*.

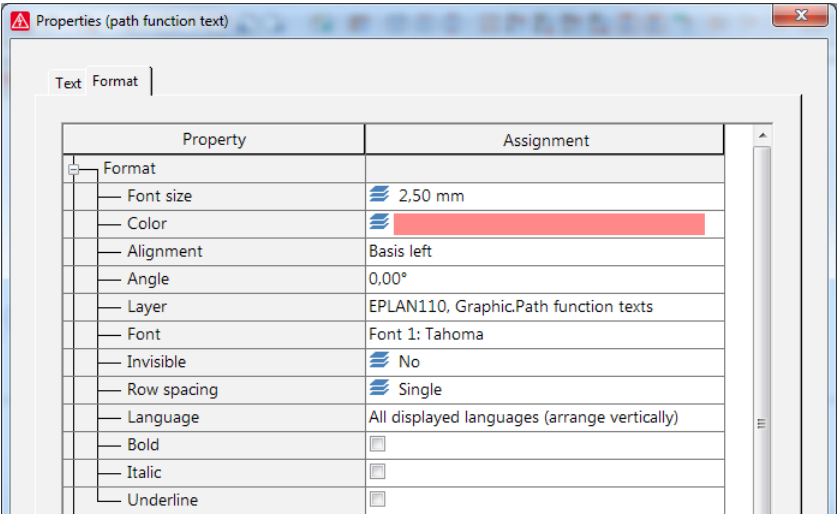


Fig. 4.130 Formatting options for texts in the Format area

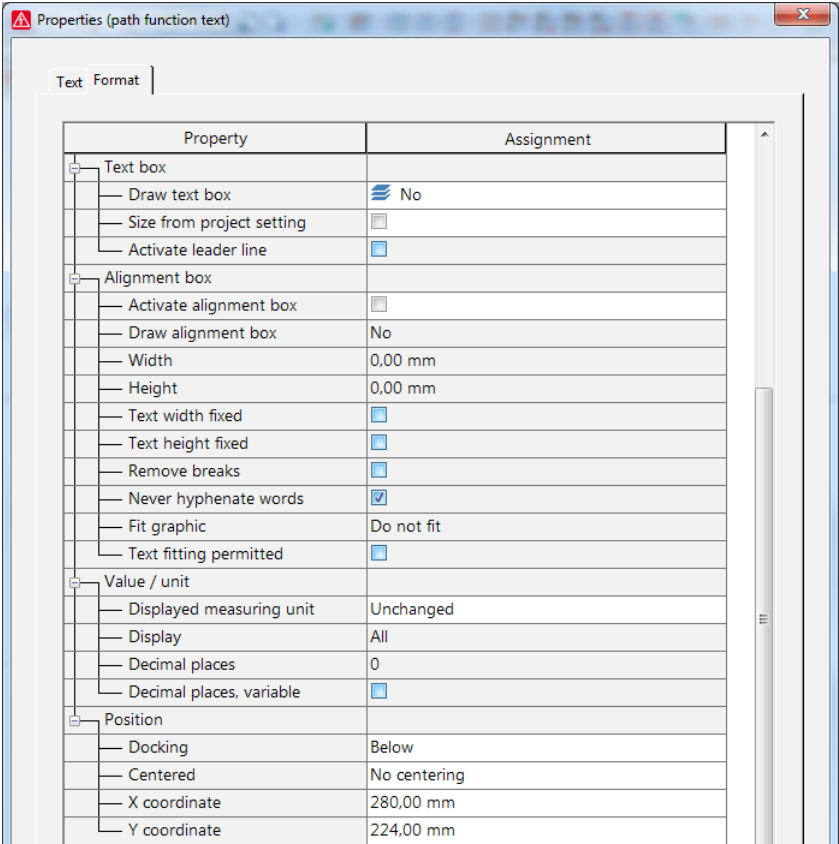


Fig. 4.131 Further options (Text box, Alignment box, Value /unit and Position)

Most fields are selection fields, meaning that when a field like the **Row spacing** field is opened, all selection options are displayed.

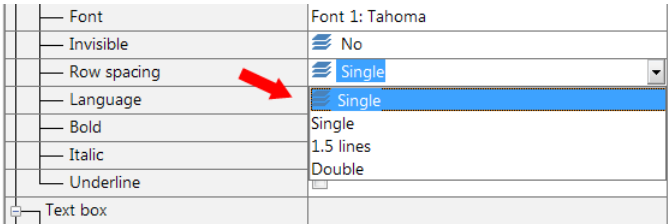


Fig. 4.132 Row spacing format property

A number of fields are also free input fields, such as the field for the **Angle** property. Here, you can select default values, such as the layer setting default for the angle, or enter your own values. If you enter your own value, thus overwriting the layer value, the symbol in front of the original entry disappears.

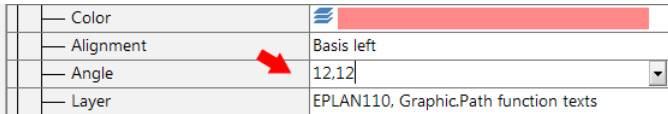


Fig. 4.133 Free input fields

The next section contains a few explanations regarding selected properties of the *Format* tab.

4.7.4.2.1 Format tab – Format node – Language

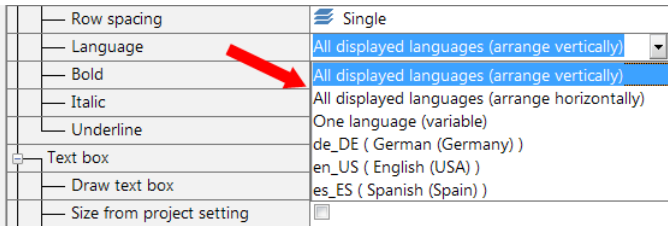


Fig. 4.134 Language selection list

The *Language* node contains the functions for displaying translated texts in the project. The **ONE LANGUAGE (VARIABLE)** selection entry depends on the settings in **UTILITIES/TRANSLATION/SETTINGS**. These settings (**SETTINGS/TRANSLATION [PROJECT NAME]** dialog) are also where the general project languages and the display languages in the project are defined.

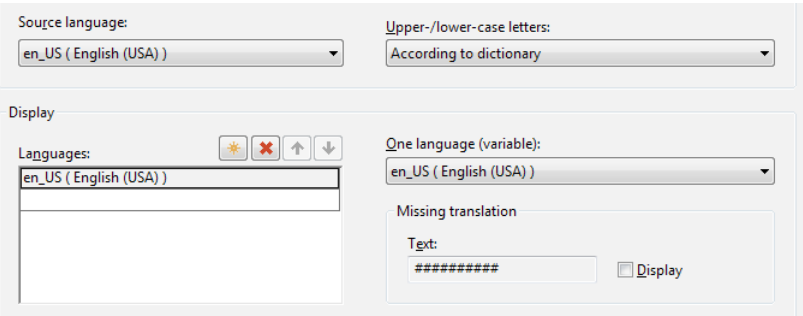


Fig. 4.135 Some translation settings

From the existing *project languages*, you can select the *displayed languages*. Depending on the existing project languages, you can then select in the *One language (variable)* selection field the desired *Displayed language*.

4.7.4.2.2 Format tab – Border node – Activate alignment box

In addition to the *Format* and *Language* nodes, there is also a *Border* node on the *Format* tab. This node contains properties, for example, to always limit the *Width* or *Height* of a

text. Other options include removing breaks from texts or defining that words (in the case of automatic breaks due to the width/height limit) must not be split.

These functions are very useful, especially for translations. This prevents translations from overlapping each other since the width of the text expansion can be limited.

Fig. 4.136 shows a typical example. In this case, it does not matter whether it is pure text or device text. The procedure is always the same.

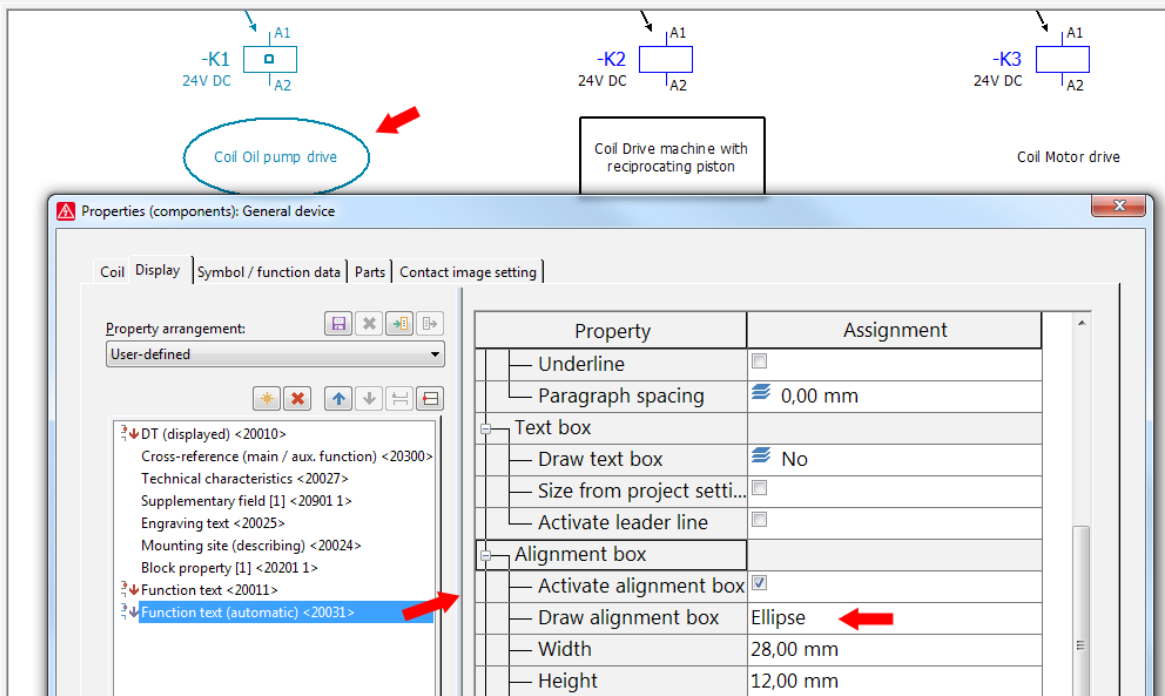


Fig. 4.136 Examples of an activated alignment box

TIP: To better read the often large number of properties and their values in dialogs, you can change the height of lines. You click the table and, while keeping the **CTRL** key pressed, turn the scroll wheel of the mouse. Depending on the direction, the lines will increase or decrease in height, improving legibility.

Property	Assignment
Format	
Font size	1,80 mm
Color	
Alignment	Middle center
Angle	0,00°
Layer	EPLAN501, Property placement.Descri...
Font	Font 1: Tahoma
Invisible	No
Row spacing	Single
Language	All displayed languages (arrange vertic...
Bold	
Italic	
Underline	
Paragraph spacing	0,00 mm
Text box	
Draw text box	No
Size from project setting	

Fig. 4.137 Large row height

Property	Assignment
Format	
Font size	1,80 mm
Color	
Alignment	Middle center
Angle	0,00°
Layer	EPLAN501, Property placement.D...
Font	Font 1: Tahoma
Invisible	No
Row spacing	Single
Language	All displayed languages (arrange ...
Bold	
Italic	
Underline	
Paragraph spacing	0,00 mm
Text box	
Draw text box	No
Size from project setting	
Activate leader line	
Alignment box	
Activate alignment box	
Draw alignment box	Ellipse
Width	28,00 mm
Height	12,00 mm

Fig. 4.138 Small row height

EPLAN remembers the settings, but handles all dialogs differently. This means that, if you wanted to, you could set different line heights everywhere.

If you want to return to the intended EPLAN row height setting, simply click **FONT SIZE 100%** in the popup menu. EPLAN then restores the default setting.

Properties (components): General device

Coil Display | Symbol / function data | Parts | Contact image setting

Property arrangement:
User-defined

DT (displayed) <20010>
Cross-reference (main / aux. function) <20300>
Technical characteristics <20027>
Supplementary field [1] <20901 1>
Engraving text <20025>
Mounting site (describing) <20024>
Block property [1] <20201 1>
Function text <20011>
Function text (automatic) <20031>

Font size

Freeze panes
Unfreeze panes
Adjust column width
Font size 100%

Property	Assignment
Format	
Font size	1,80 mm
Color	
Alignment	Middle center
Angle	0,00°
Layer	EPLAN501, Prop
Font	Font 1: Tahoma

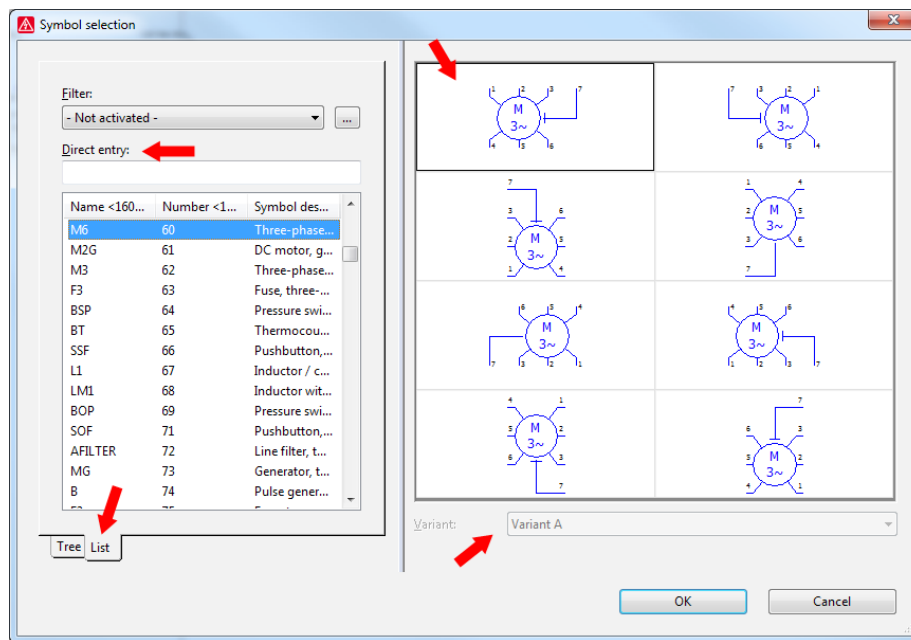
Fig. 4.139 Restore default font size 100 % setting

■ 4.8 Components (symbols)

Components, also called symbols, are among the most important elements in EPLAN and, naturally, of any schematic. Logical connections and reports can be generated based on components and their entered properties or assigned function definitions

4.8.1 Insert components (symbols)

Symbols can be inserted in EPLAN in different ways. The usual way is via the **INSERT/ SYMBOL** menu or the **INS** key. EPLAN then opens the **SYMBOL SELECTION** dialog. In this dialog, you can select the appropriate symbol. You can either use direct entry (*List view*), in which case you must have a certain amount of knowledge of identifiers, and enter the symbol names directly into the **Direct entry** field, or you can select the symbol selected from the list using the mouse.



Symbol selection(list)
dialog

Fig. 4.140 Symbol selection via list view

Next to the symbol list is the *Symbol preview*. EPLAN (generally) provides symbols with eight variants. The normal four variants are Variant A = 0°, variant B = 90°, variant C = 180° and variant D = 270°. The remaining four variants E to H are 180° mirrored variants of the first four variants.

As with many other dialogs, it is possible to set a filter in this dialog via the  selection button. You can use the filters to display only *Multi-line symbols* or only a special *Symbol library*.

For example, the *Only symbols of the same symbol representation type* parameter can be used to display only multi-line symbols from all displayed symbol libraries.

In addition to the *List view*, the **SYMBOL SELECTION** dialog also lets you select symbols from a *tree view*. In contrast to the list view, symbols here are listed according to a structure. Depending on the application, a symbol can be found more quickly when you search, for example, for the *motor overload switch* symbol using plain text.

Symbol selection (tree)
dialog

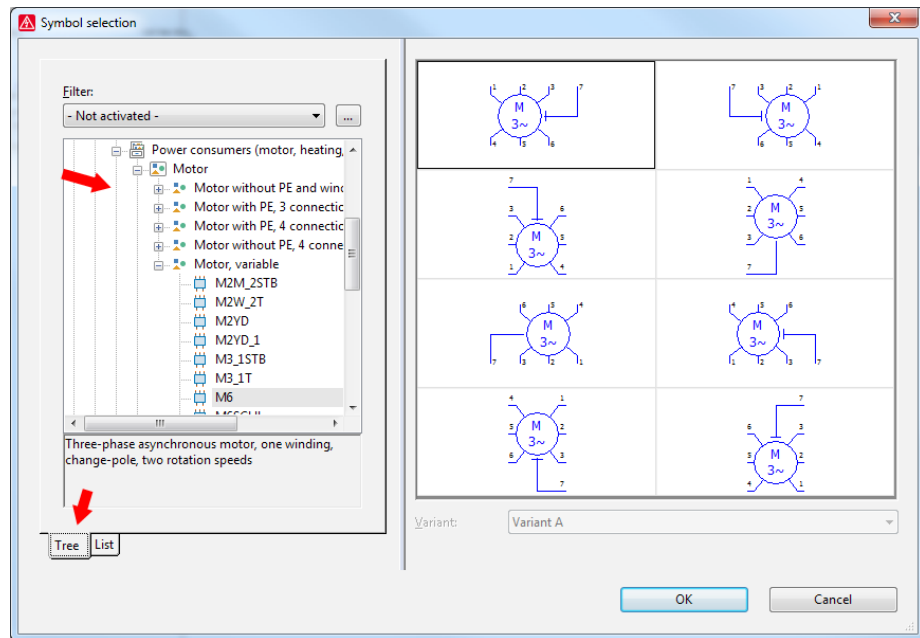


Fig. 4.141 Symbol selection via tree view

Once you have selected a component (symbol), it can be applied to the schematic with **ENTER** or the **OK** button.

4.8.2 Properties (components) dialog – [device] tab

The symbol that has been applied “hangs” on the cursor and can be placed in the schematic.

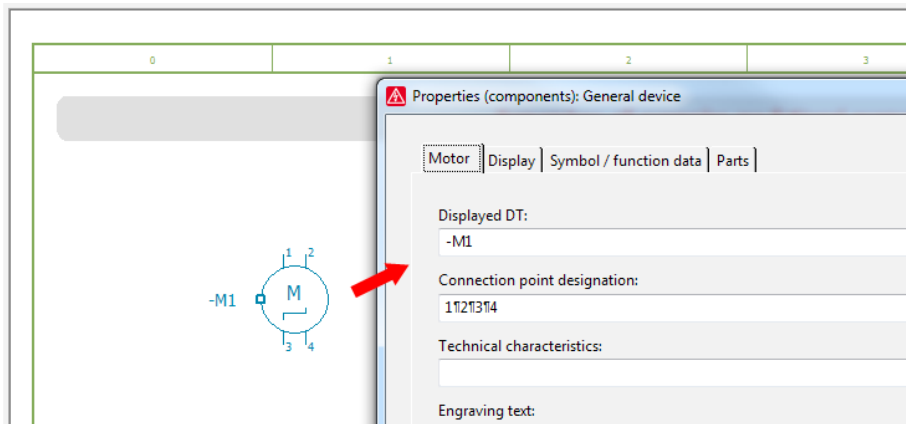


Fig. 4.142 Properties (components) dialog

When you place the symbol, EPLAN opens the **PROPERTIES (COMPONENTS) [DEVICE TYPE]** dialog. The dialog consists of the different tabs *[Device type]*, *Display*, *Symbol / Function data* and *Parts* (in case of a main function).

Here is where you find fields for the **Displayed DT** (can be edited), the **Full DT** field (can be edited indirectly via the  selection button and the follow-up **Full DT** dialog where the DT is split into its elements, which can be edited individually), the **Connection point designation** (selectable from the selection list), **Connection point description** (selectable from the selection list), **Technical characteristics**, **Function text** (of the symbol!) fields, the **Engraving text** field (can be edited and selected from a selection list if there are already other entries in the project for this) and the **Mounting site (describing)** field, which can also be selected from a selection list.

Aside from these items, there is also an important check box: *Main function*. This box lets you decide whether this symbol is to be a main function (can carry a part), or an auxiliary function (does not contain parts data).

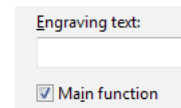


Fig. 4.143
Main function

The lower half of the **PROPERTIES (COMPONENTS) [DEVICE TYPE]** dialog consists of the **Property name** and **Value** columns (of the properties). The  button can be used to add new properties.

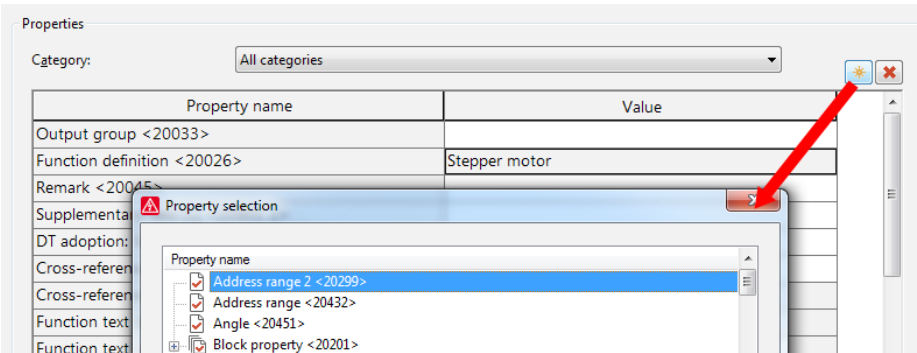


Fig. 4.144 Add new properties

When you click the  button, EPLAN opens the **PROPERTY SELECTION** dialog. Here, you can select a new property and then click **OK** to apply it in the **PROPERTIES (COMPONENTS)** dialog. All properties that are not gray in the **Value** column of this dialog can be changed.

4.8.3 Display tab

The *Display* tab allows you to add new symbol properties to the existing *for display* default symbol properties at the symbol, or delete existing properties.

The *Display* tab also contains all elements for changing the display of the properties. On this tab, you can change the format, font size or font for each property, including the connection points of devices. In this case, you would open the *Connection points* tab in the property arrangement.

The toolbar can be used to insert new properties, move the sequence of properties, or dock and undock them.

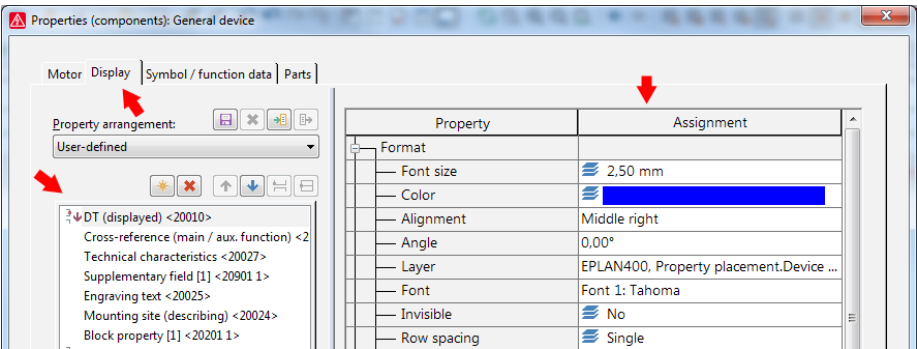


Fig. 4.145 Display tab and its options

This section explains *docking and undocking of properties* in more detail. *Docked* properties are those that do not have a symbol in front of the property name. These properties are assigned to the next higher property that has a symbol.

In Fig. 4.146, all properties framed in red would make up a “block”. If you move one of the properties out of this “block”, all the other properties in this “block” will be moved as well.

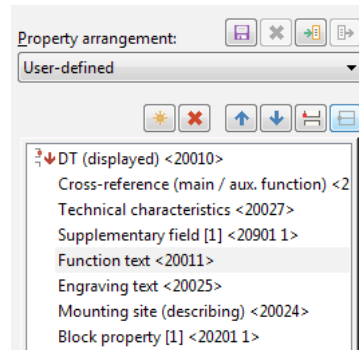


Fig. 4.146
Related properties

If you undock a property, it will be removed from the “block” and is now free. Now it is independent of the other properties and can be placed freely.

To undock a property, you select it and use the  button to undock it. Then, using the direction buttons, you move it to the end of the property arrangement. If you do not do this, the undocked property and all other properties that follow it in the display will form another “block”.

The *Display* tab also contains all the functions you need to format or position text. From placing a border around text to setting a different font size for a specific property or defining/adjusting a position (X/Y).

The procedure is the same as with texts and will not be discussed further here. The control of the language display will also not be discussed again.

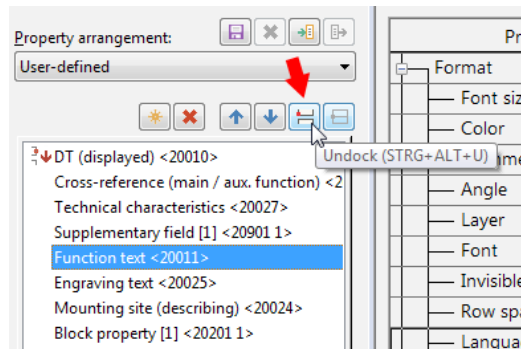


Fig. 4.147
Undocking a property

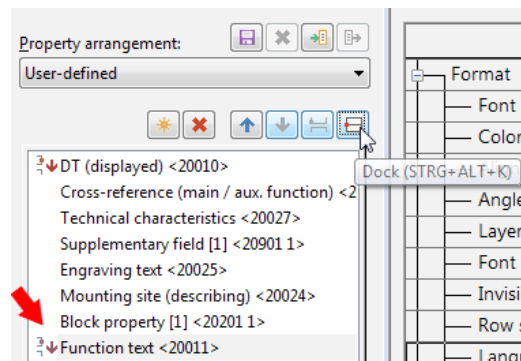


Fig. 4.148
Undocked and moved property

4.8.3.1 Activate leader line property

There are some differences from formatting text when it comes to certain properties. One example is the *Activate leader line* setting.

The *Activate leader line* setting allows you to activate a leader line for each property displayed on a device. This way, for example, you can undock or move properties, while the leader line ensures a type of “connection” with the actual device.

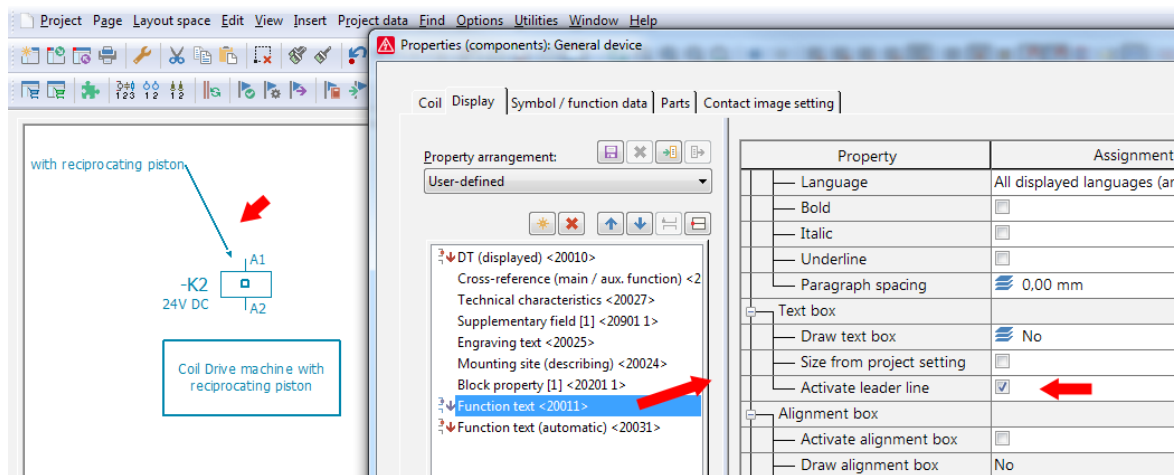


Fig. 4.149 Example of activated leader line

4.8.4 Symbol/function data tab

The *Symbol/Function data* tab contains important information for assigning **logic** to the graphical symbol.

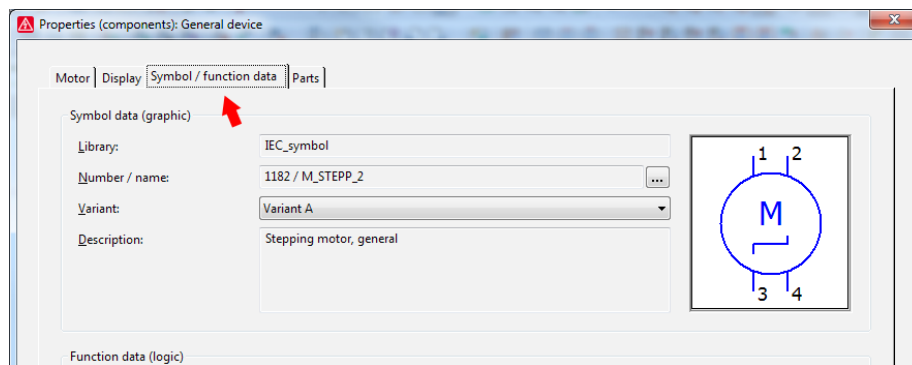


Fig. 4.150 Graphical information for the symbol

The upper area contains the graphical symbol data. In the **Number/Name** field, the  button can be used to exchange the existing symbol for a different one. A different angle variant of the symbol can be selected in the **Variant** field. You simply open the selection list and select the desired variant. All other fields (**Library**, etc.) are derived from the graphical symbol data and cannot be edited.

The lower area of this tab contains the logical symbol information, such as which *Function definition* is attached to the symbol, the *Representation type*, or whether it is a **MAIN FUNCTION**, etc.

Function data (logic)

Category: Electrical engineering: Motor

Group: Motor without PE, 4 connection points

Definition: Stepper motor

Description: Stepper motor, 4 connection points (1+2 => Coil 1, 3+4 => Coil 2)

Connection points: 4

Representation type: Multi-line <1>

☒ Main function
☐ Safety function
☐ Intrinsically safe
☒ With signal isolation
☐ Net-connecting

Logic...

Fig. 4.151 Logical symbol functions

In the **Definition** field, the button can be used to assign a different function definition to the symbol if necessary. All other fields are then derived automatically from the applied function definition.

Connection points: 4

Representation type: Multi-line <1>

- Multi-line <1>
- Multi-line <1>
- Overview <3>
- Pair cross-reference <-2>
- Single-line <2>
- P&ID diagram <38>
- Functional <-12>
- External <-3>
- Graphic <7>

Fig. 4.152 Representation type selection

The **Representation type** here can be modified from a selection list. This may be necessary, for example, if you wish to create or represent an overview on a multi-line schematic page.

Aside from the previous graphical and functional information, there are additional settings options under the **LOGIC** button. In a targeted manner, you can change or adjust logical information such as **Connection point type** or the **Number of targets**, **Potential type** and much more for the individual connection point of a device.

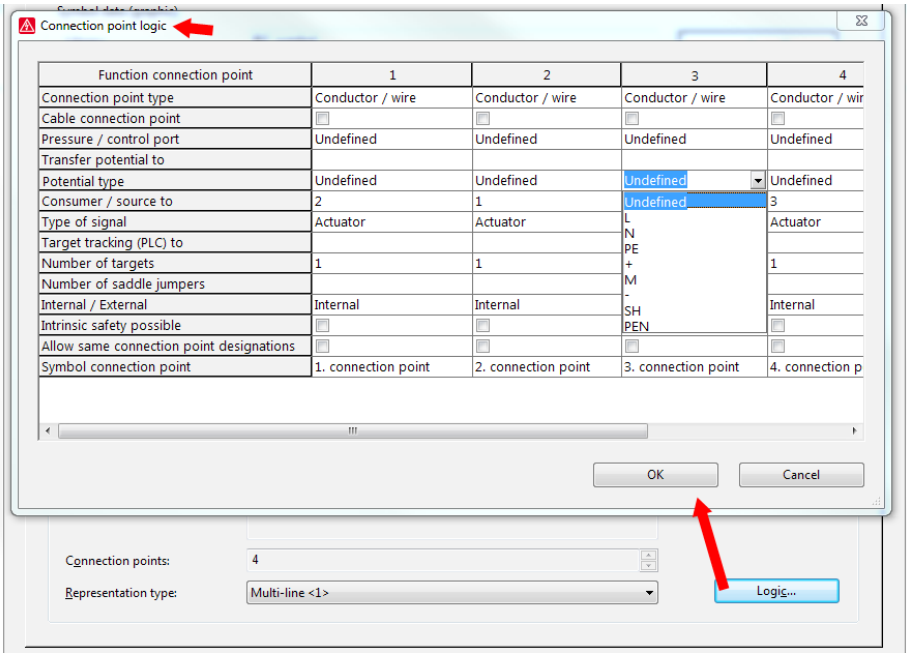


Fig. 4.153 Logical functions of individual connection points

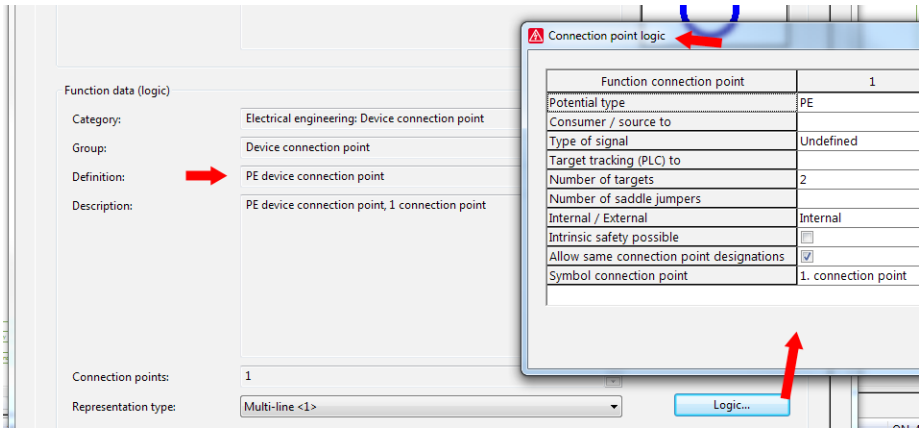


Fig. 4.154 Allow same designation

A typical example of changing the *Connection point logic* is the enabling of the *Allow same connection point designation* setting when using device connection points (here PE). This makes it possible, like with terminals, to use the same designation several times (PE rail), and the check run will not find any duplicate connection point designations.

4.8.5 Parts tab

On the *Parts* tab, you can assign one or more parts to a symbol. Click the **Part number** field and then use the [...] button that appears to switch to parts management.

In **parts management**, you select the necessary part and click OK to apply it to the symbol. Parts management automatically closes after the part is applied.

In addition to simple part selection, this dialog also has a **DEVICE SELECTION** button. In contrast to simple part selection, this will offer only devices for selection that fit the existing functions of the symbol in the project. This ensures that EPLAN selects, for example, a coil for a coil rather than a motor overload switch. **DEVICE SELECTION** is the preferred method.

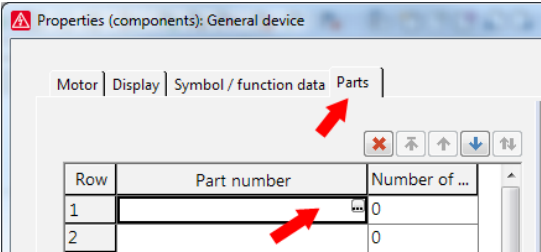


Fig. 4.155 Simple part selection

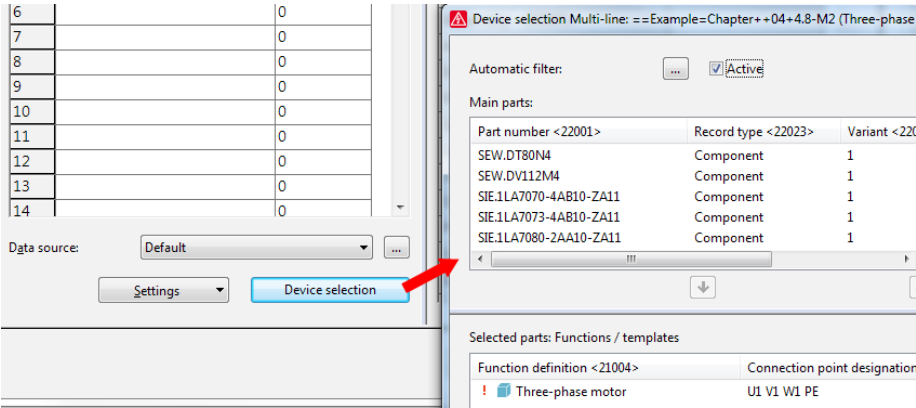


Fig. 4.156 Device selection

■ 4.9 Cross-references

Cross-references are one of the most important elements in EPLAN. When automatically generated, they create visual and logical connections, e.g. between *main and auxiliary functions*.



NOTE: Inserting and filling out of the corresponding symbols for the cross-reference display examples will not be explained at this point. More details are available in the following sections.

4.9.1 Contact image on component

For example, a motor overload switch or a pushbutton switch can have auxiliary contacts. The auxiliary contacts physically belong to the **motor overload switch** or **pushbutton switch** symbol and, for the sake of clarity, should also be displayed at the switch.



In the following, a motor overload switch is inserted into the schematic. The motor overload switch initially has no auxiliary contact at the symbol. As such, this is not a problem, because EPLAN creates a report on the auxiliary contacts used in the schematic on the basis of the settings on the *Display* tab in the *Contact image* selection field (the selection here is set to **ON COMPONENT**). This way, the contacts used and their cross-references are displayed automatically.

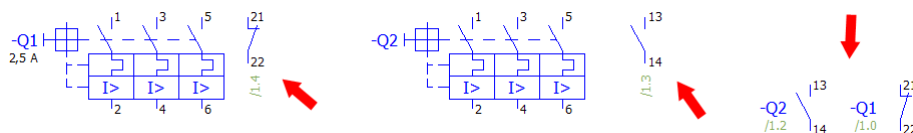


Fig. 4.157 Automatic cross-referencing

This does not matter provided that both or all auxiliary contacts that have such a motor overload switch are also used in the schematic.

Not all auxiliary contacts are always used, and EPLAN would display the motor overload switch only with the auxiliary contacts used (as graphical representation at the motor overload switch). The other auxiliary contacts physically exist but are “suppressed” with this setting. To display those contacts anyway, a device with the appropriate *Function definition* must be assigned to the **Motor overload switch** symbol.

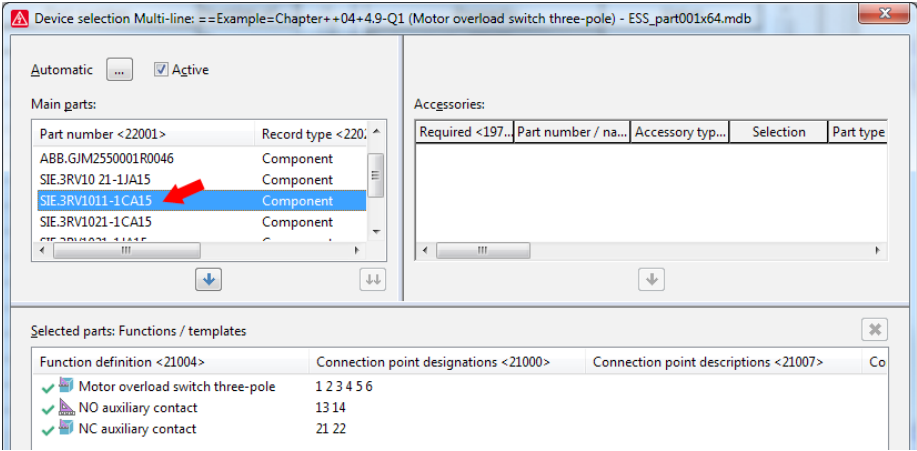


Fig. 4.158 New device selection

This is best done using the Device selection. You open the properties of the symbol, switch to the **Parts** tab and click the **DEVICE SELECTION** button.



TIP: The **SETTINGS** button provides access to the settings defining how EPLAN should behave during device selection: whether existing function data is used or not; whether any other criteria of devices or functions are be taken into account, etc. All these have to be taken into consideration during device selection.

This is important to know because in certain situations this can result in devices not being displayed in the device selection. In situations such as “incorrect” or inappropriate settings in the device selection, or when there is no part with the appropriate functions in parts management.

Back to device selection: EPLAN recognizes that the motor overload switch consists of the motor overload switch itself and a placed auxiliary contact (considered a function). Now all parts that fit the selected motor overload switch (including its used functions), i.e. the parts with their function definitions, are offered for device selection.

In the upper area of the **DEVICE SELECTION FUNCTION** dialog, you can now select the right part and click  to apply it. EPLAN “moves” the part across the existing functions of the motor overload switch and adds the functions not yet placed to the function template.

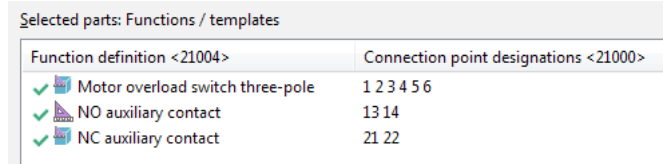


Fig. 4.159 Selected part with its function template

The **DEVICE SELECTION FUNCTION** dialog can now be closed with the **OK** button. EPLAN transfers the part into the *Parts* tab. If you now leave the **SYMBOL PROPERTIES** dialog by clicking the **OK** button, EPLAN will use the stored function definitions to graphically recreate the entire contact image for the auxiliary contacts. It makes no difference whether the auxiliary contacts have been used in the schematic or not.

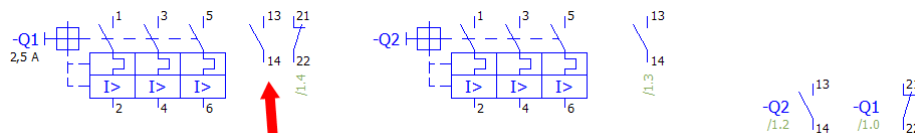


Fig. 4.160 Complete contact image

But this representation can be achieved only if the display of the contact image has been set accordingly in the symbol properties of the *Display* tab.

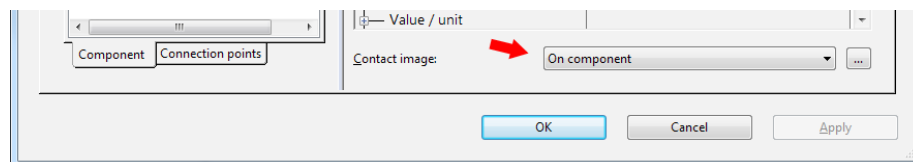


Fig. 4.161 Contact image setting

This display of the placed contacts is set by EPLAN automatically at first. But if you wish to, or must, deviate from this automatic setting, you can modify the item manually. You simply click the **MORE** button. In the **CONTACT IMAGE POSITION** dialog that follows you can manually edit the position for the Y and X values.

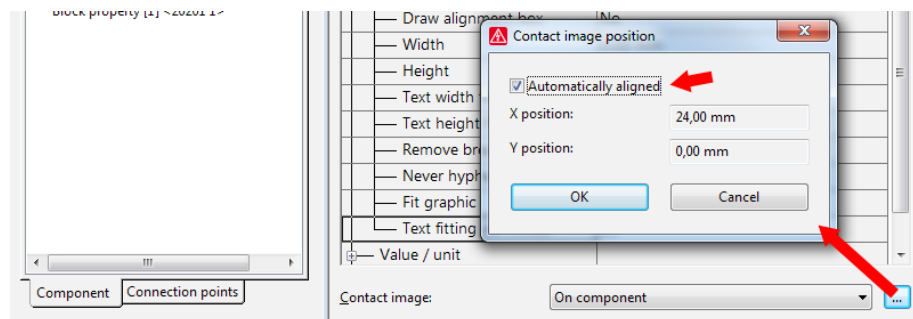


Fig. 4.162 Manual editing of the contact image position

4.9.2 Contact image in path

In contrast to a contact image at the component, which is usually displayed to the right of the symbol, for (e.g.) contactors the contact image is generated by EPLAN below the contactor coil in the column (path).

Similar to the previous example, at first only the contacts used are automatically displayed in the contact image.

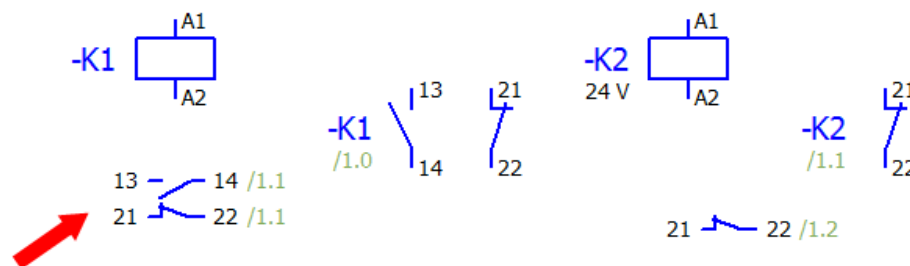


Fig. 4.163 Automatically generated contact image of the functions in use

To obtain a complete *contact image*, **DEVICE SELECTION** is performed in the same way as was done in the example of the motor overload switch. The procedure is identical to that already described in the motor overload switch example. After device selection, the full contact image is displayed. If this does not occur, then the *Contact image* entry in the **DISPLAY** tab of the *Symbol properties* dialog must be checked. It must be set to the **In path** entry.

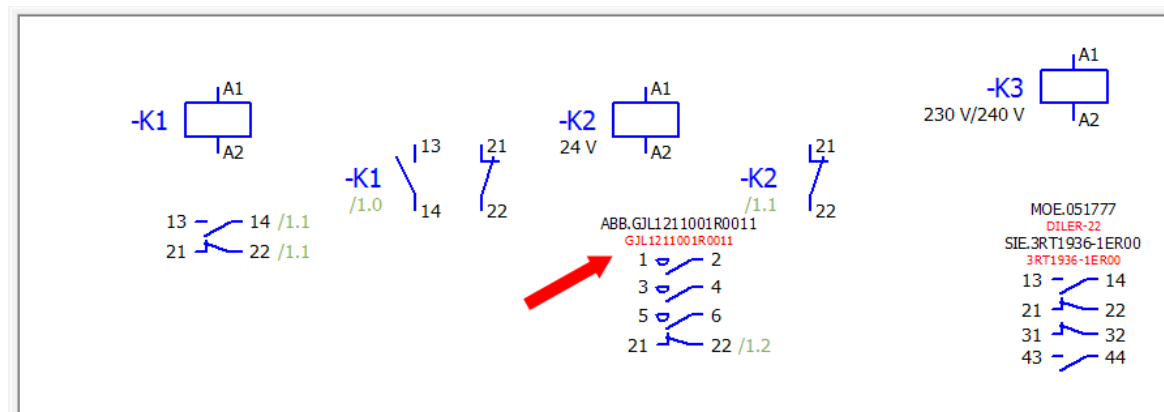


Fig. 4.164 Complete contact image following device selection



NOTE: EPLAN also lets you move contact images. Every contact image can be individually moved (it makes no difference whether the contact image type is *On component* or *In path*).

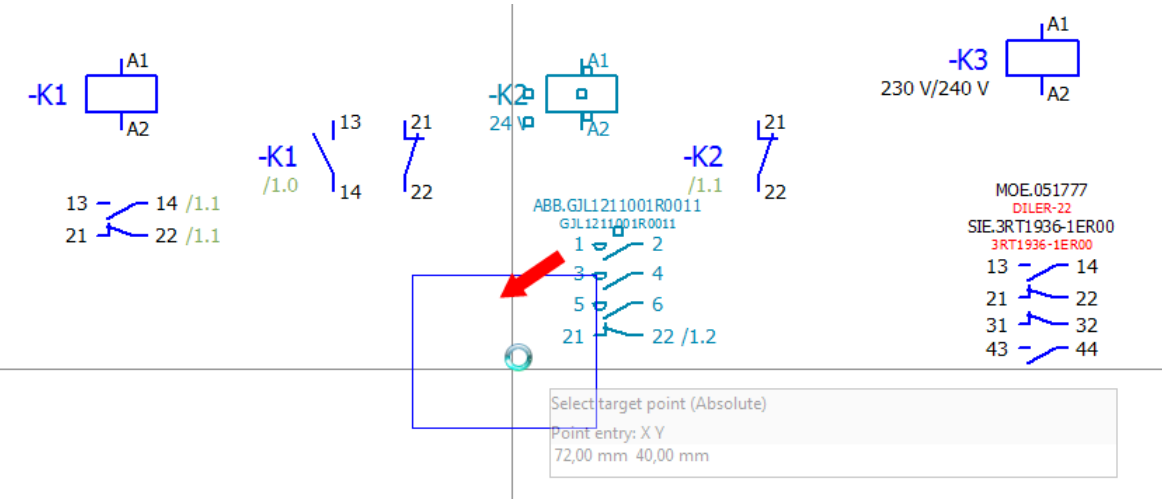


Fig. 4.165 Moving the contact image manually

You first select the symbol, and then press **CTRL + B** or use the **EDIT/TEXTS** menu to call the **MOVE PROPERTY TEXTS** function. EPLAN then displays a small blue rectangle for “grabbing” with the mouse. You then click the contact image. It will then “hang” as a blue rectangle on the cursor and can be placed as desired (while keeping left mouse button pressed). You left click to place it in the desired position.

At this point, it should be mentioned that you can define the positions of the contact images for the motor overload switch and/or contactor coils in the respective plot frame (or only for this page, in which case, in the Page properties), for example, as **Contact image margin (on component) <12059>** and **Contact image margin (in path) <12060>** plot frame properties.

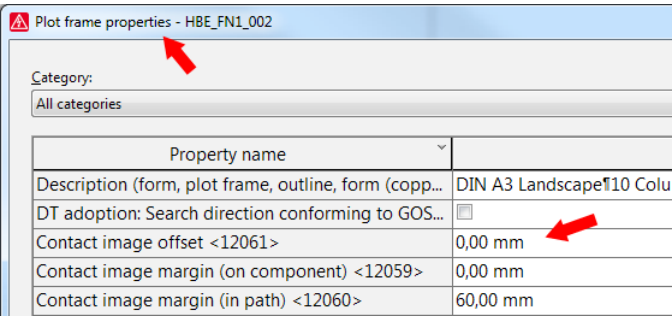


Fig. 4.166 Properties of contact image offsets

Regardless of these default values, all contact images can, of course, be individually and manually set with the desired values and placed at the respective devices in the schematic.

4.9.3 Special feature: Pair cross-reference

Aside from the previous examples, the functionalities of EPLAN have been used on the basis of the device selection and its functions. But situations can arise where there are no parts with correct function templates. It is, after all, only because of the function templates that EPLAN knows, for example, that an illuminated pushbutton consists of several functions.

Using an illuminated pushbutton as an example, we will now see how contact images can be referenced to the interconnected functions even without parts. This is done using the **Pair cross-reference** representation type (in the **PROPERTIES (COMPONENTS)** dialog on the *Symbol/Function data* tab in the **Representation type** selection list).

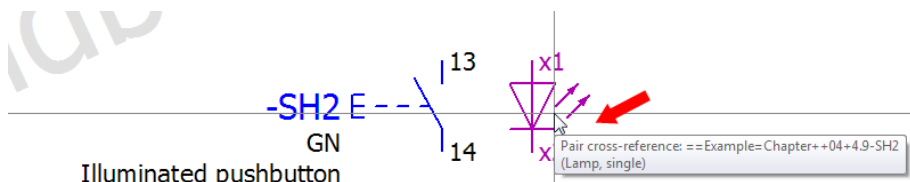


Fig. 4.167 Insert and designate symbols

A pushbutton is inserted. It is assigned a DT and is designated as a main function. In addition to the pushbutton, a lamp is also inserted. It is **not assigned a DT**, nor is it **designated as a main function**. The representation type of the lamp is set to **Pair cross-reference**.

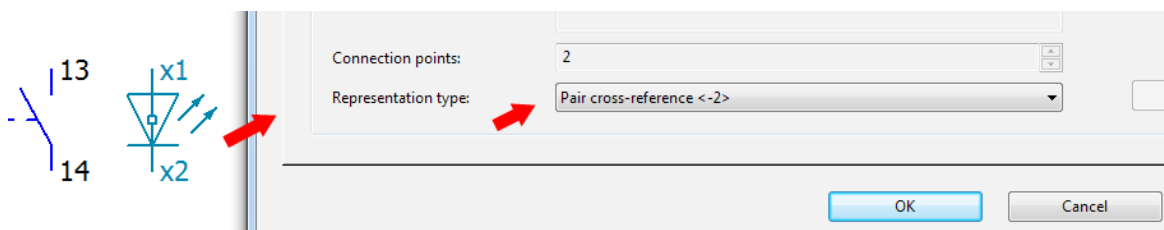


Fig. 4.168 Setting the pair cross-reference representation type

Now, at the auxiliary function (of the lamp), the **Cross-reference (main/aux. function)** **<20300>** property is set to the optimal position of the display of the cross-reference. The following settings are made: the **angle to 90°**, the **alignment to center right**, the **position** of the **X coordinate to 0.00 mm** and of the **Y coordinate to -6.00 mm**.

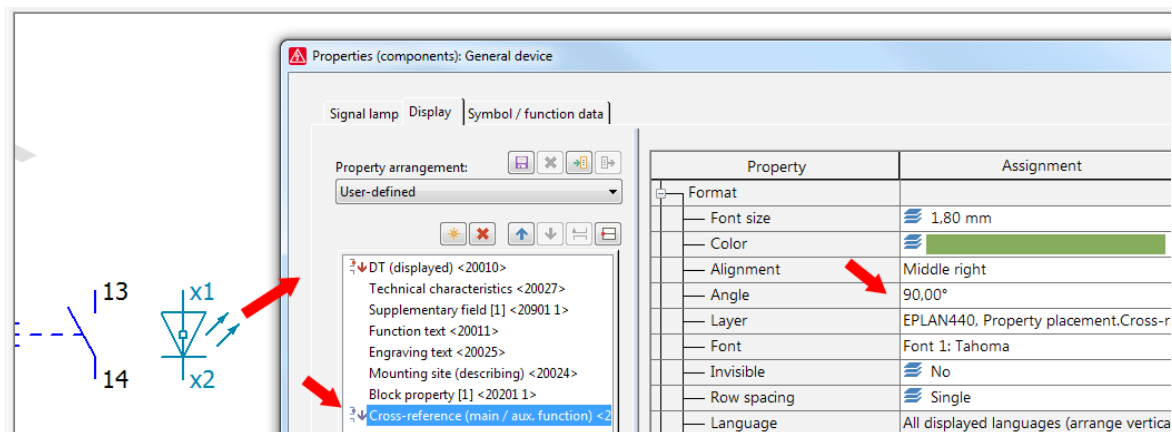


Fig. 4.169 Setting appropriate values for the cross-reference

If all values have been set accordingly, the cross-reference will be rotated subsequently by 90° directly below the lamp.

Now the lamp can be placed, either from the **device navigator** via the **PLACE** popup menu or via the symbol selection.

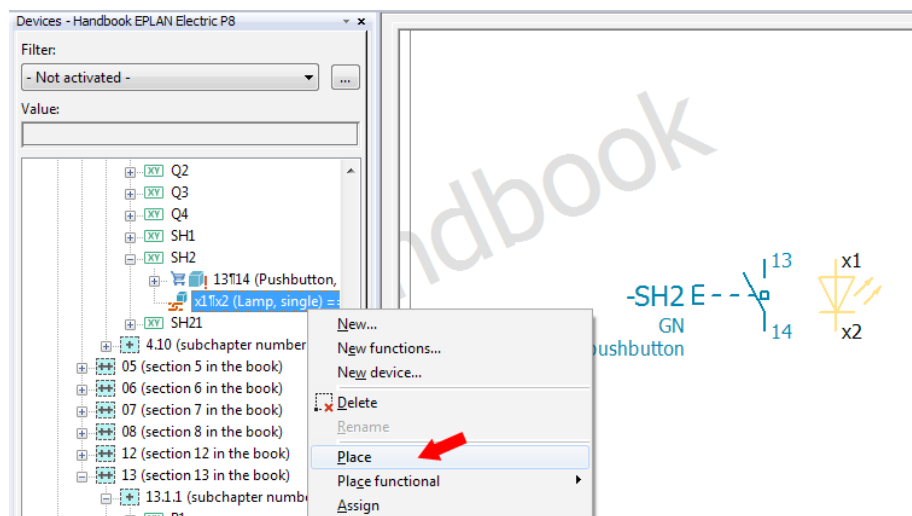


Fig. 4.170 Place multi-line function from the device navigator

After this function has been placed, EPLAN immediately displays the cross-reference on both pages: on the complete device illuminated pushbutton and on the device of the lamp itself, which is displayed as distributed.

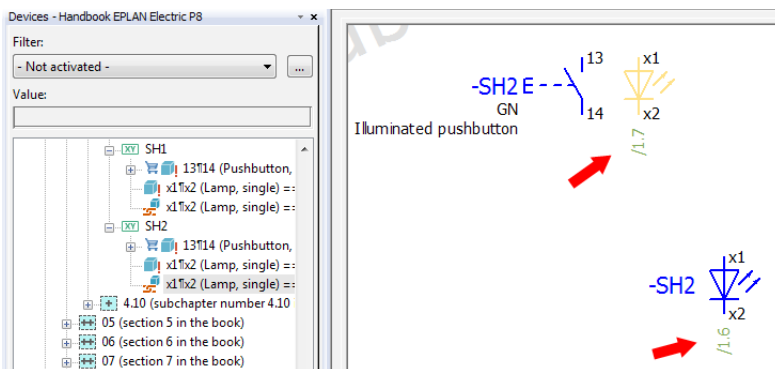


Fig. 4.171 Cross-reference with pair cross-reference representation type

Finally, the default property arrangement setting on the display of the distributed symbol is set. This puts the cross-reference in the “right” position.

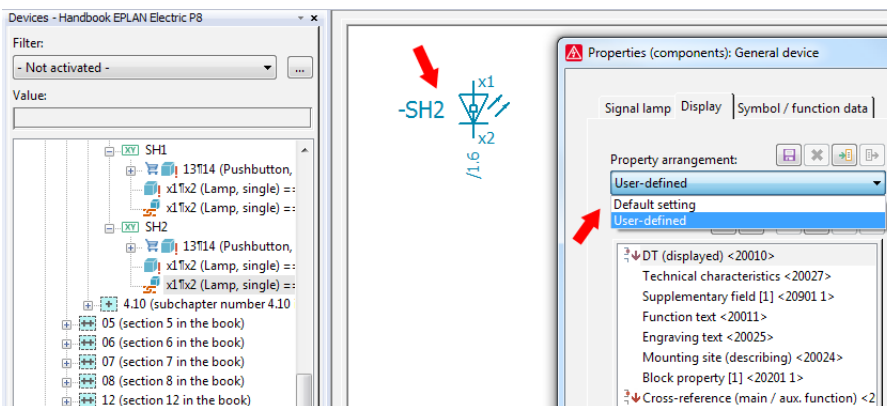


Fig. 4.172 Default property arrangement setting

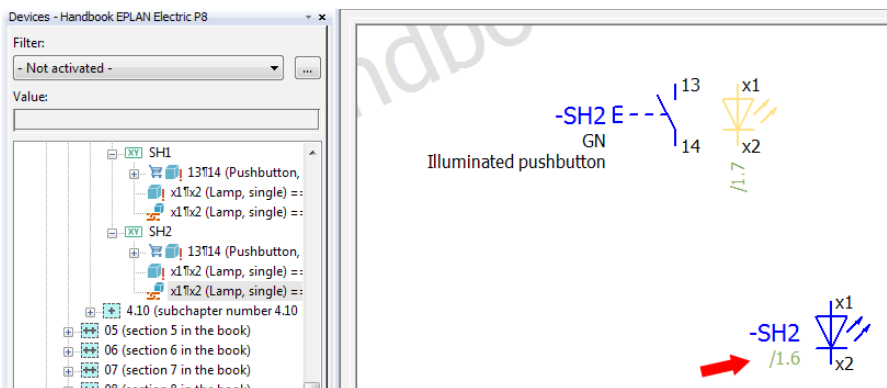


Fig. 4.173 Result



NOTE: The placing of functions from the device navigator is always preferable to manual placing (insert symbol, fill out DT, etc.), because EPLAN implements all key settings automatically, which means that everything fits all the time.

4.9.4 Contact image distributed device list

Aside from the usual methods of creating contact image types (in the path or on the component), EPLAN offers a further variant of contact images. Contact image based on a special form, distributed device list *.f45.

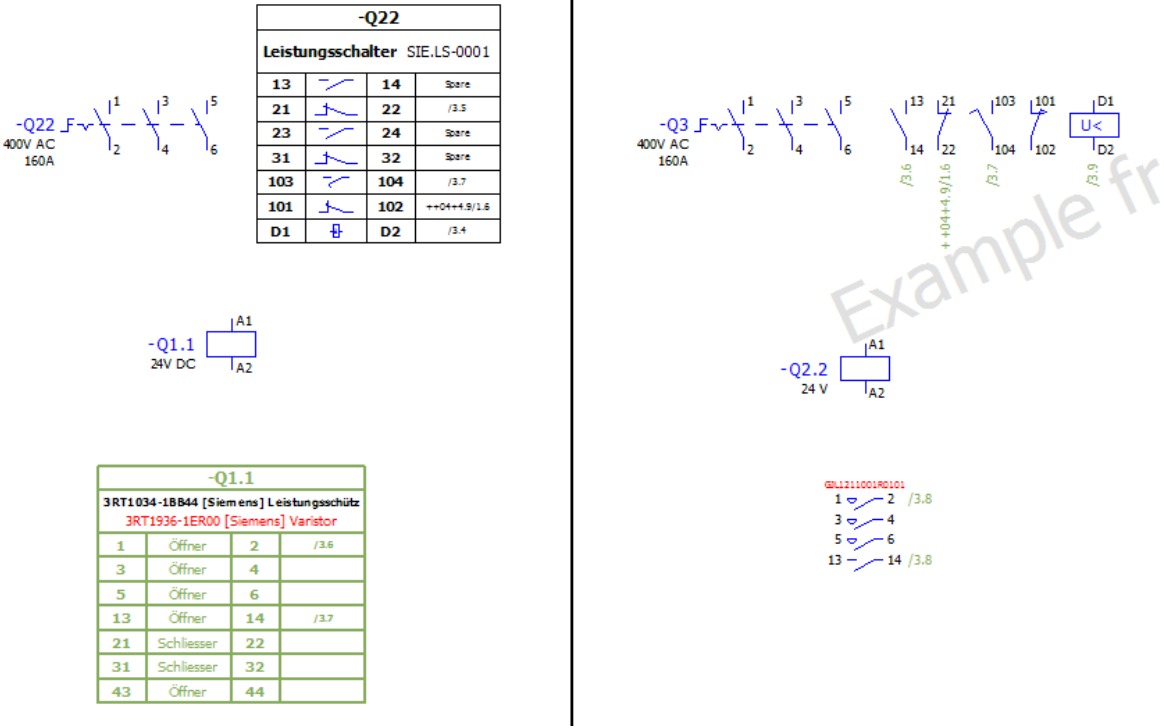


Fig. 4.174 Examples of form-based contact images

These contact images are based on the form distributed device list *.f45. The use of this type of contact images can be set directly in the symbol properties. For this, open the *Contact image setting* tab and select the **EMBEDDED FORM** option.

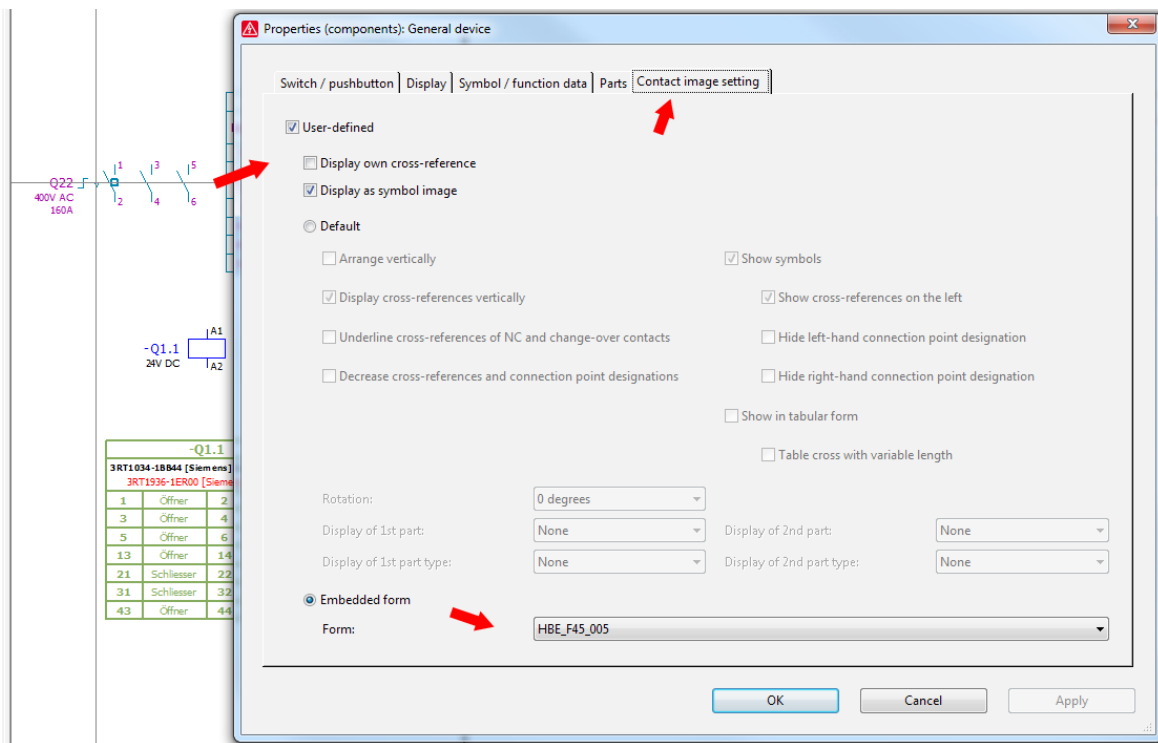


Fig. 4.175 Contact image setting tab

Now EPLAN allows you to select a form from the **FORM SELECTION LIST**, or if you click **BROWSE**, EPLAN opens the **SELECT FORM** dialog.

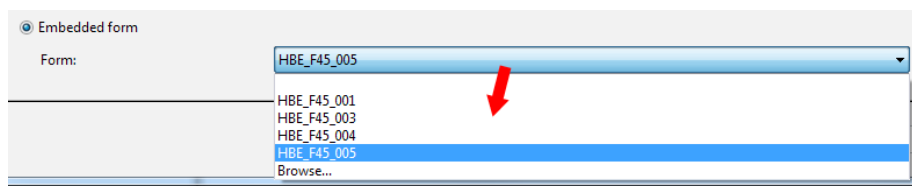


Fig. 4.176 Select form

The dialog can be closed after the selection. EPLAN now realizes the form with the desired information. Of course, the position of the contact image can be modified in any way you choose, as is also possible with the other types of contact image display.



NOTE: If there is no **CONTACT IMAGE SETTING** tab in the **PROPERTIES (COMPONENTS): (...)** dialog, you must use the **Display** tab (lower right) to set the display of the contact image from **NONE** to **ON COMPONENT** or **IN PATH**.

■ 4.10 Device selection settings

There are a number of settings for the device selection. Depending on how they are set or adjusted, they can influence **DEVICE SELECTION** and thus the selection of parts.

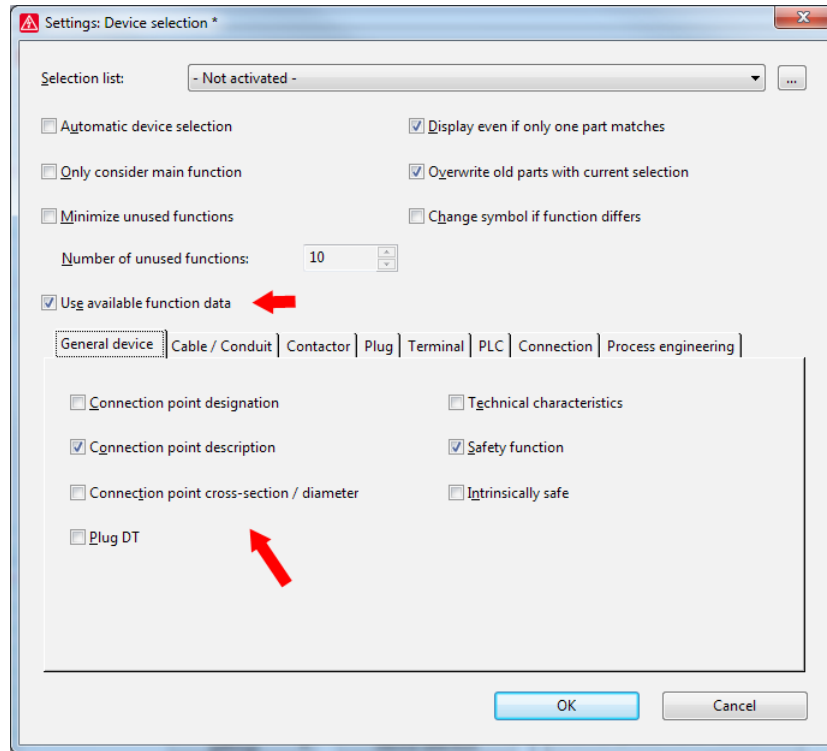


Fig. 4.177 Device selection settings

It helps if the *Use available function data* setting and the desired properties have been enabled. This way, only parts will be offered for selection that match the identifying function data. Identifying function data is the data that is located on the tabs, e.g. *General devices*, *Cable*, *Contactor*, etc.

If the *Use available function data* setting is disabled, you can select devices freely. This means that the function data available in the project (at the function) does not have to match the function data in the parts selection.

5

Navigators

The graphical editor with its aids, such as the navigators, is the central focus of project editing. Whatever approach you use, whether graphically-oriented, object-oriented, or a mixture of both, you always remain in the graphical editor. Only the way in which you use it can differ.

What is a navigator? A navigator provides a particular “viewpoint” for looking at project data. As the name indicates, the cable navigator only ‘looks’ at devices with the *Cable* function. The PLC navigator only looks at devices with the *PLC* function.

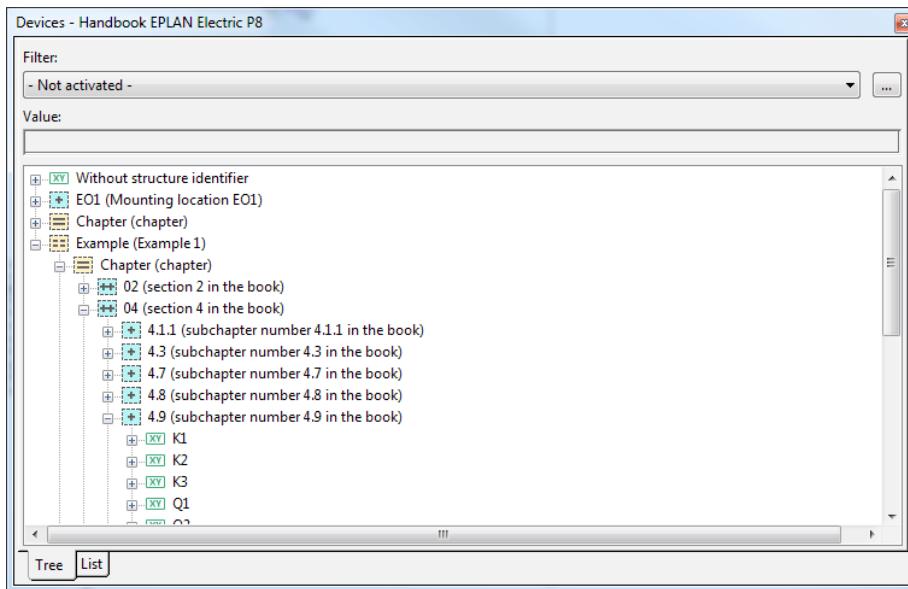


Fig. 5.1 A navigator and its view of devices

This chapter covers navigators, their basic functions and the additional functions in the **PROJECT DATA** menu. Numerous explanatory examples are used to demonstrate the correct use of devices, their settings and their properties.

To work quickly and efficiently with EPLAN, you must make use of the various navigators because they allow certain project actions to be performed much faster and more effectively – the key phrase here is **bulk editing**.

This chapter will not deal with all navigator functions, nor will it deal with all of the navigators in EPLAN Electric P8. The essential purpose of this chapter is rather to explain functions that are often used in daily work.

■ 5.1 Overview of the most important navigators

When possible, it is a good idea to keep some navigators open all the time, for example the **device navigator** (this is the most comprehensive navigator as it displays all of the project data).

But the device navigator is not the only navigator. EPLAN has a separate navigator for each of the different **device types** in the project data. Each of these navigators has specific tasks and features relating to the selected project data.

In addition to this, every **PROJECT DATA/[DEVICE TYPE]** menu item contains extra functions with a functional scope similar to the navigator menus. Some of the EPLAN navigators are listed here.

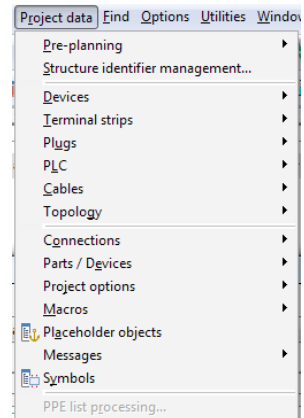


Fig. 5.2
Navigator variety

The **Device navigator** shows the view of the *project data* and can, for example, directly edit a device's properties. But it cannot, for example, number terminals; this can only be performed by the special terminal strip navigator.

The **Terminal strip navigator** only shows project data for *terminals*. It has terminal editing functions such as renumbering and creating terminal strips.

The **Plug navigator** only shows project data for *Plugs/Sockets*. It has functions for editing plugs and sockets.

The **PLC navigator** only contains project data relating to *PLC* functions such as, for example, creating new PLC cards. It provides an overview of other PLC box functions (these can also be frequency converters with an integrated EA layer). The PLC navigator provides different views (representations) of the project data.

The **Cable navigator** contains all project data for *cables* and their properties. With the cable navigator, you can, for example, place or collect *conductors*.

The **Topology navigator** contains information pertinent to the *Routing path networks* (2D method), such as routes and routing points.

The **Connection navigator** contains all project data related to the *connections* in a project. You can use this navigator, for example, to use existing connections.

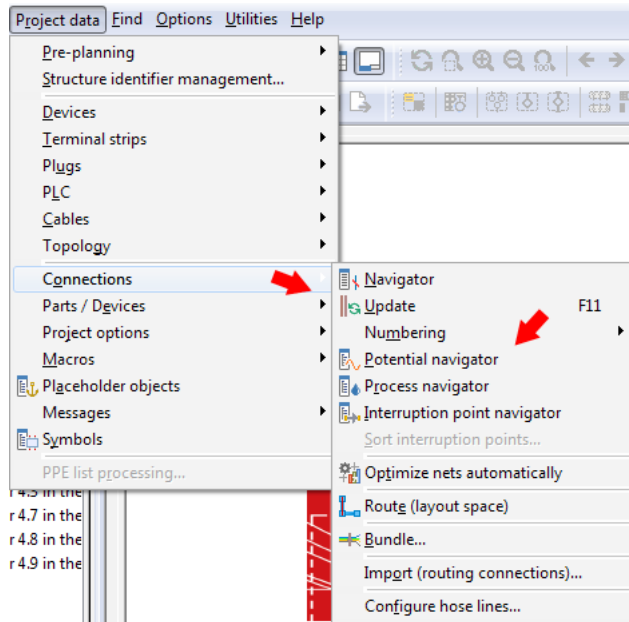


Fig. 5.3 Additional navigators for connections project data

The connections project data also includes the **Potential navigator**, the **Process navigator** and the **Interruption point navigator**. These navigators manage potentials, signals, networks, interruption points and the processes defined in the project.

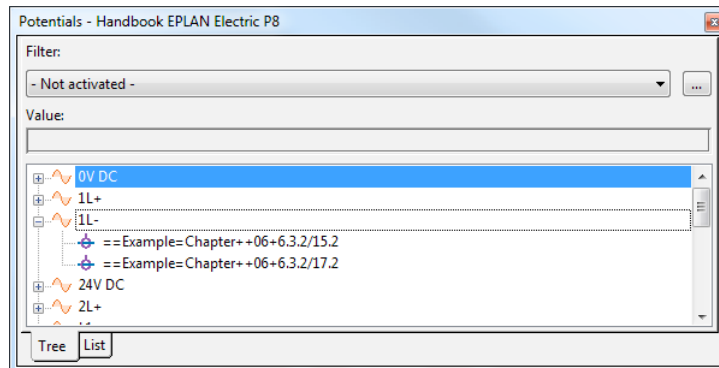


Fig. 5.4 Potential navigator

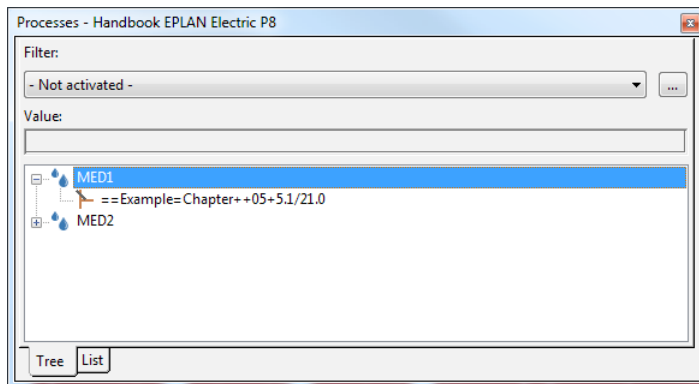


Fig. 5.5 Process navigator

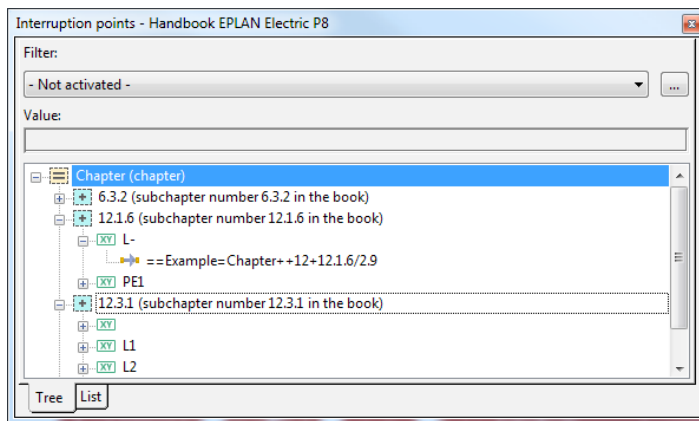


Fig. 5.6 Interruption point navigator

There are also additional navigators in the **PROJECT DATA/PARTS/DEVICES** menu. The **bill of materials navigator** contains all functions for centrally editing parts, such as swapping parts or selecting a contactor. Unlike the other navigators, the bill of materials navigator also provides special views of the project data in a tree view.

There are also two more navigators in the **PARTS/DEVICES** menu.

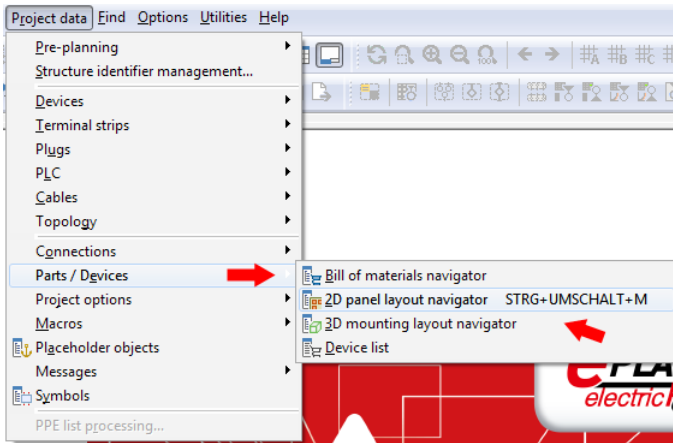


Fig. 5.7 Navigators in the parts / devices menu

The **2D panel layout navigator** allows, for example, the creation of a 2D structure diagram of the devices used in the project, based on technical data (at least width and height) from the parts master data or manually entered data.

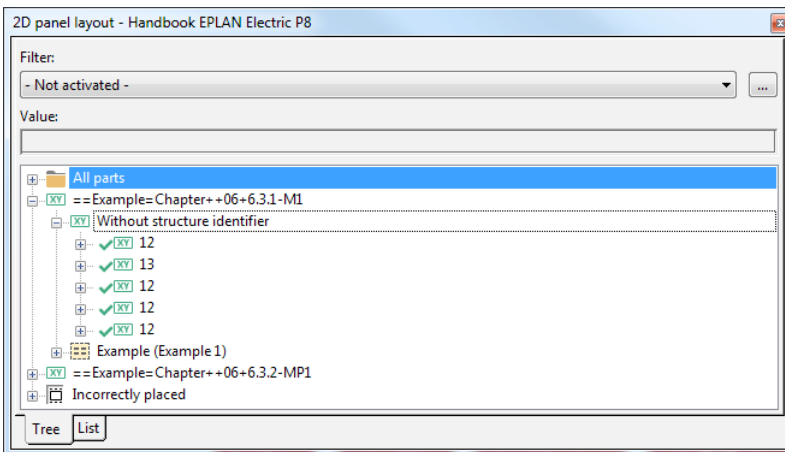


Fig. 5.8 2D panel layout navigator

The **3D mounting layout navigator** allows a 3D diagram of the devices used in the project, based on technical data (at least width, height and depth) from the parts master data.

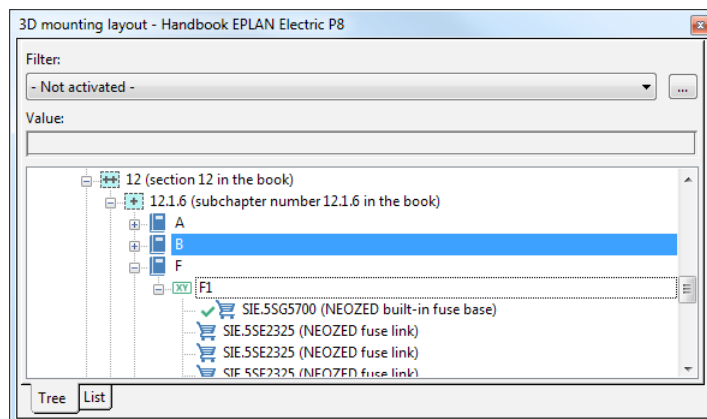


Fig. 5.9 3D mounting layout navigator

The **Project options navigator** allows creation of schematics using various options. The options can be easily switched on and off, by page, across pages, or in page sections.

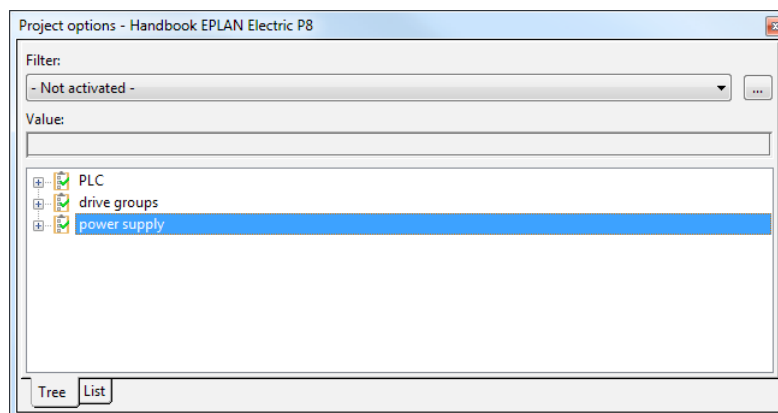


Fig. 5.10 Project options navigator

The **macro navigator** displays and manages conveniently macros of all types (page, window, symbol or 3D macros) in one single macro project.

Note: So far, the navigator serves no other functions in a schematic project.

The **Placeholder objects navigator** conveniently manages macros with value sets from a single location, from changing global value sets editing selected objects.

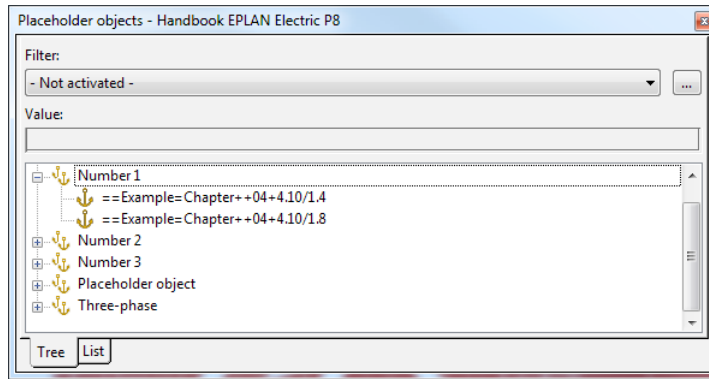


Fig. 5.11 Placeholder objects navigator

5.1.1 Additional navigators and modules

The following entry is not a navigator in the sense of those under section 5.1: **Symbol selection** in the PROJECT DATA/SYMBOLS menu. This entry allows symbol selection to stay open all the time like a navigator.

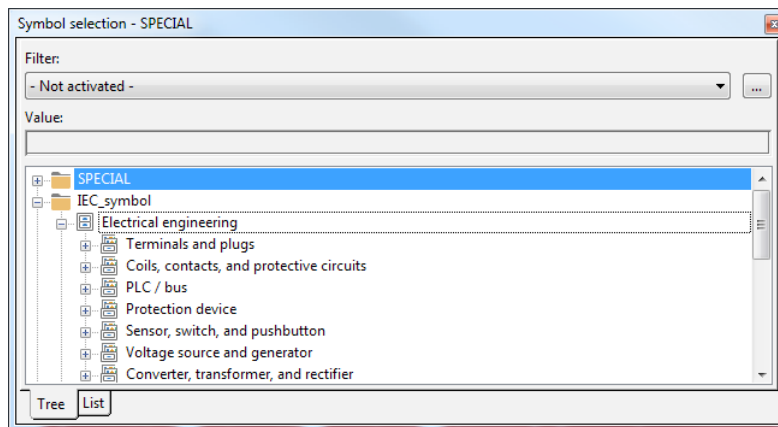


Fig. 5.12 Permanently open symbol selection

The following navigators are found at other locations: the **Part master data navigator** and the **EPLAN Data portal navigator**. The part master data navigator is started in the UTILITIES/PARTS/PART MASTER DATA NAVIGATOR menu, and the EPLAN Data Portal navigator is started in the UTILITIES/DATA PORTAL menu.

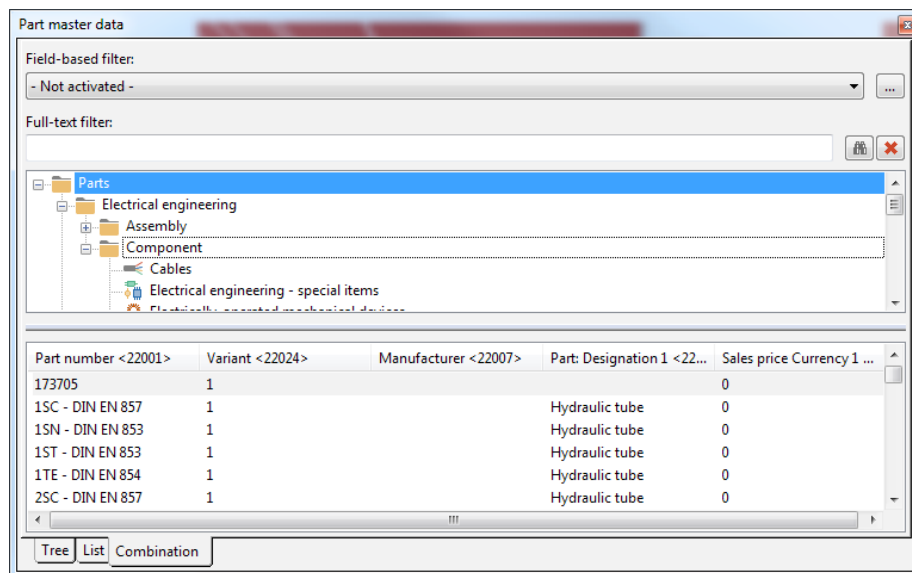


Fig. 5.13 Part master data navigator

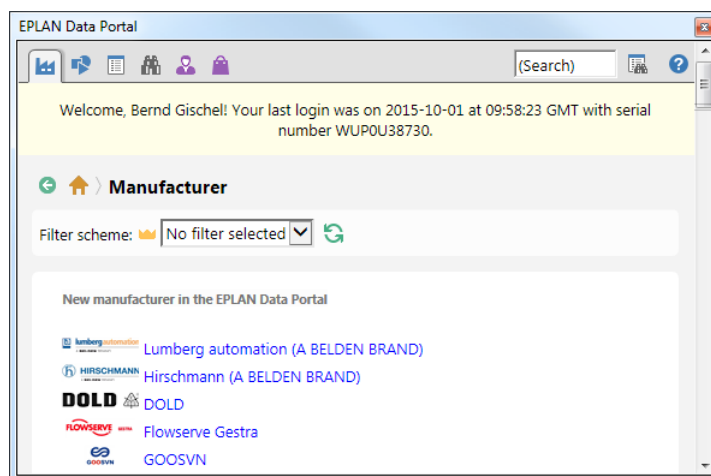


Fig. 5.14 EPLAN Data Portal

These navigators are independent dialogs that can remain open in the same way as 'true' navigators. For example, the **Parts master data navigator** always allows access to parts management without having to open parts management via UTILITIES/PARTS/ADMINISTRATION.

In addition, the PAGE menu also contains the COMMENT NAVIGATOR, which displays and manages entries made externally in the PDF documentation.

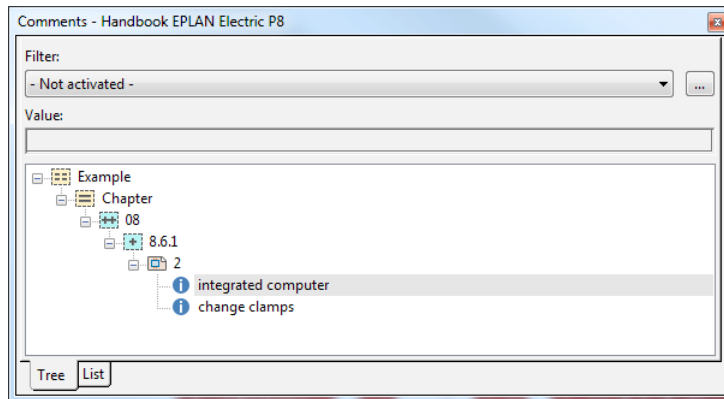


Fig. 5.15 Comment navigator

One should also mention other navigators in passing, which can be used in connection with the add-ons (sold separately) in the **PROJECT DATA** menu.

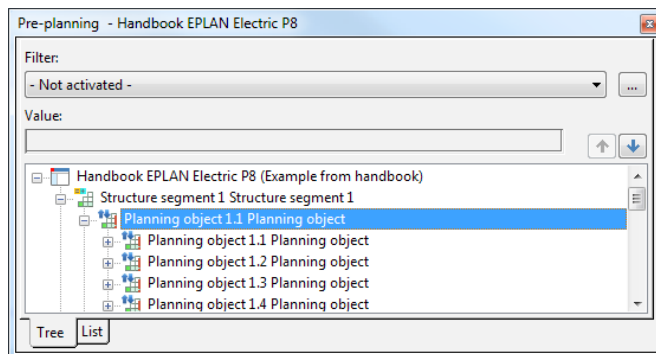


Fig. 5.16 Pre-planning navigator

The same menu also contains the **Segment template navigator**.

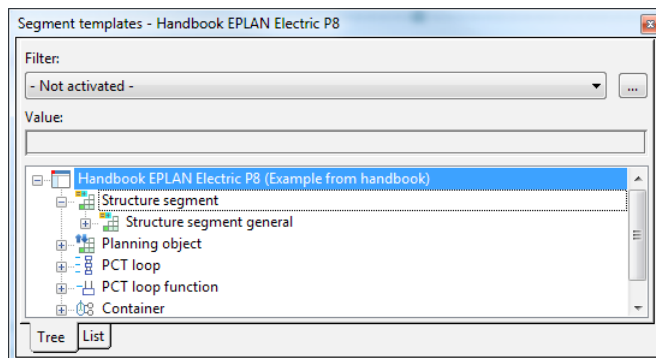


Fig. 5.17 Segment template navigator

Apart from these navigators for the pre-planning of projects, the **PROJECT DATA** menu also contains the **Topology navigator**.

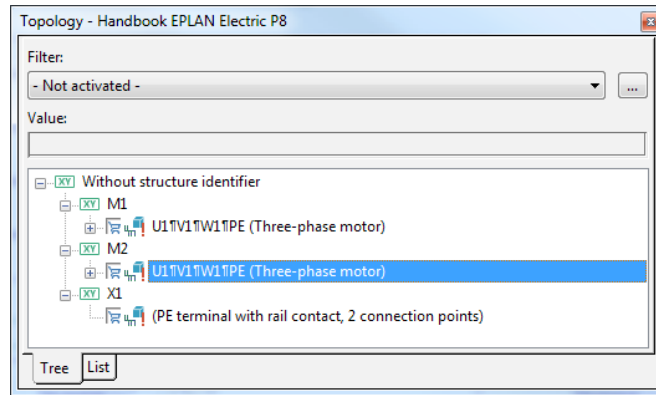


Fig. 5.18 Topology navigator

5.1.2 General navigator functions



NOTE: Functions that are identical or very similar in terms of the approach in the respective navigators are described in section 5.9.

This concerns the following functions in the popup menu of the respective navigator:

- Place
- Assign
- Filter and filter with quick entry

■ 5.2 Device navigator

All information and project data come together in the **device navigator**. The device navigator can be accessed via the **PROJECT DATA/DEVICES/NAVIGATOR** menu item, and can then be positioned anywhere on the desktop. The device navigator can do everything that can be manually done to a device during device editing, i.e. changing device tags, changing the display format, editing a device's properties, and much more. But the major advantage of the device navigator is that it does everything from a central point while maintaining a view of the entire project.

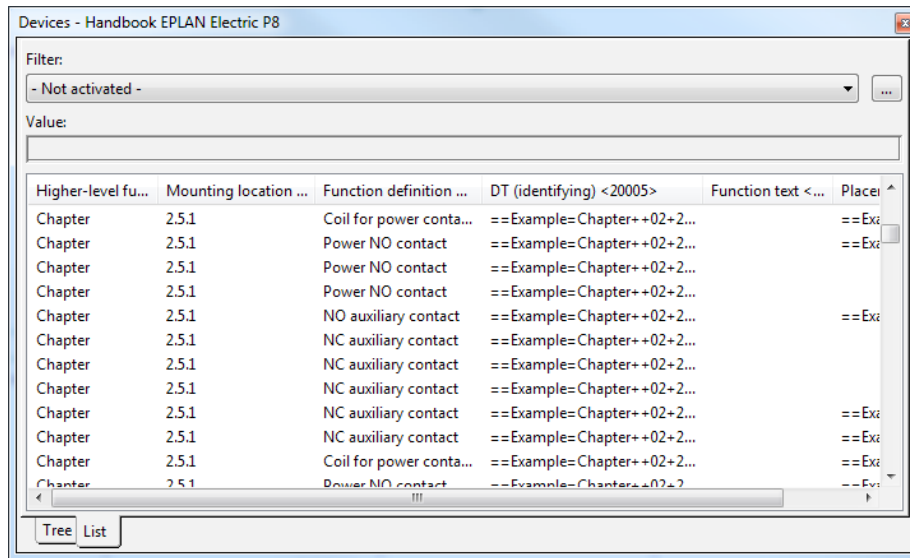


Fig. 5.19 List view in the device navigator

The *filter* and *sorting features* of the device navigator can be used to swap device navigator function definitions or connection point designations simply and conveniently. The device navigator can do more than just display and edit the properties of a device. You can also use it to edit or number several devices at once via the block functions (bulk editing).

This section not only deals with the device navigator and its functional scope, but also with a series of other functions in the **PROJECT DATA/DEVICES** menu. For reasons of space or because they are more or less self-explanatory, this chapter does not provide an overview of all of the menu items and their functions.

For the device navigator, it is a good idea to create your own keyboard shortcut for opening and closing the navigator, since it is used most often and must be opened and closed constantly (unless it remains open all the time anyway). My recommendation is **CTRL + SHIFT + D** for the device navigator and to always keep it and the page navigator open.

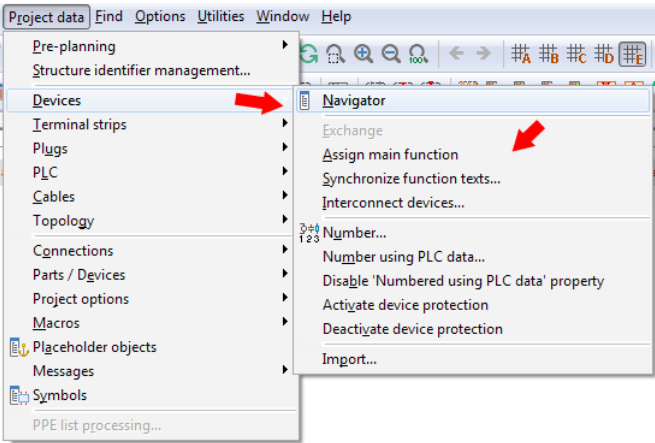


Fig. 5.20
Shortcut key in the
device navigator

5.2.1 Swap

This function allows to swap the device tags (the DT) for two selected devices. It is easy: the two devices in question are selected (it makes no difference whether this involves the main function or auxiliary function), and then the **SWAP** command is executed. Now EPLAN swaps out both DTs for all functions affected.

5.2.2 Assign main function

EPLAN distinguishes between a **main function**, for example a coil, and an **auxiliary function**, such as the contacts belonging to a coil. The *main function* setting on the first tab in the **PROPERTIES (COMPONENTS)** dialog, or on the *Symbol/function data* tab, turns a device into a main function.

Fig. 5.21
Main function setting
in the Properties
(components) dialog

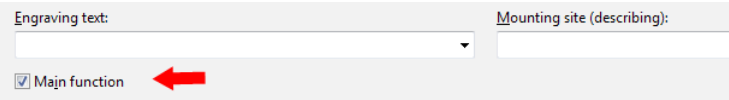
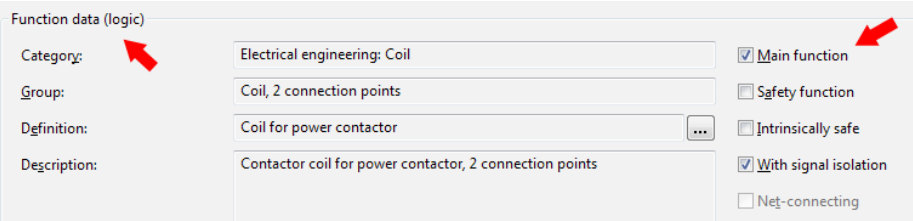


Fig. 5.22
Main function setting
on the Symbol/
function data tab



This is very important because, for example, only devices that have a main function can have parts stored on them or have devices selected for them. If devices do not have a main function characteristic then, for example, they cannot be numbered.

In general, therefore, every independent item should initially be a main function. However, if other functions with the same device tag exist in the project, then these are forced to become auxiliary functions.



NOTE: Only one main function per device is allowed at all times. Double main functions are not allowed and are detected with a check run and reported (if the corresponding check run message is used).

In the **PROPERTIES (COMPONENTS)** dialog, both ways of activating a main function are always set when one of them is set. Regardless of where you set the check box for the main function, the other input check box is also automatically activated or deactivated.

The advantage of the **ASSIGN MAIN FUNCTION** menu function here is that the main function need not be manually set for every device where this is necessary. It is entirely possible that copy operations or other similar actions can cause a device to have no main function.



NOTE: Usually, EPLAN automatically checks the main function checkbox after inserting a new symbol.

In the example, all of the devices on the schematic page were selected in order to renumber them. As an alternative, of course, these devices can also be selected in the device navigator.

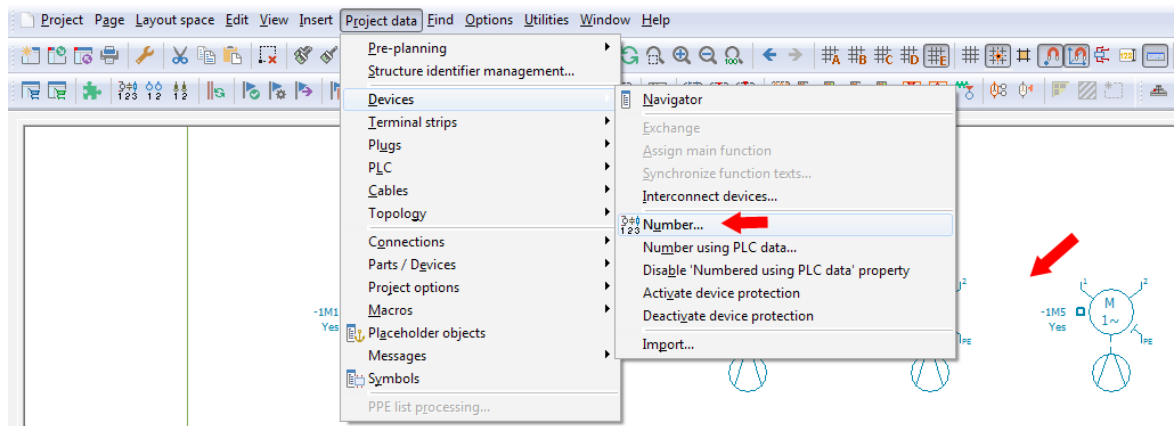


Fig. 5.23 Selected devices

After they were selected, the **NUMBER** function in the **PROJECT DATA/DEVICES/NUMBER** menu was executed. EPLAN, however, will not want to number all of the devices. Some of the devices will not be numbered because they do not have the **main function** characteristic.

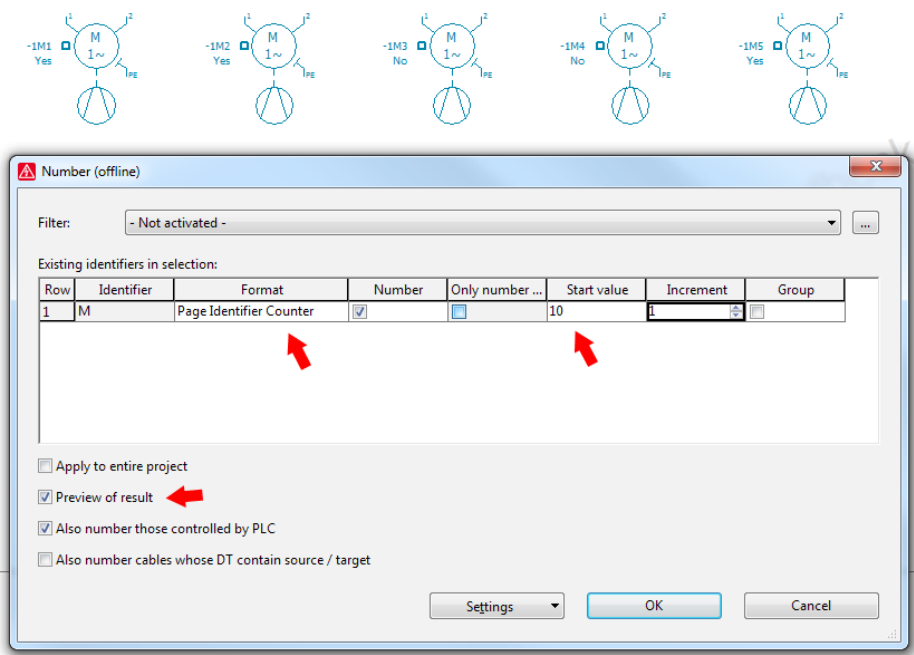


Fig. 5.24 Definitions before numbering

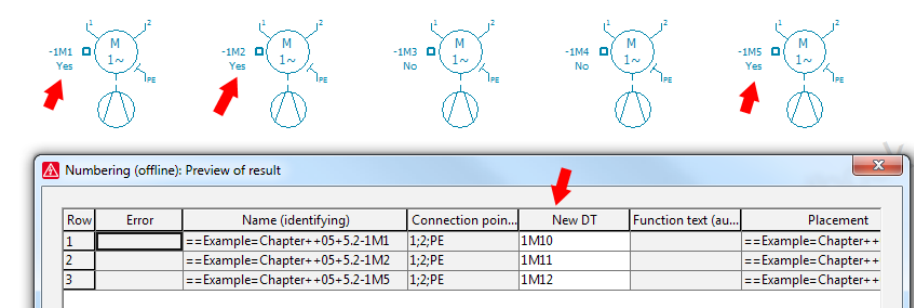


Fig. 5.25 Preview of result before final numbering

 **TIP:** I recommend wherever possible to display the preview of result in the follow-up dialogs. Even if this may take a moment longer, this preview is advantageous because you can cancel immediately if necessary without something “happening” to the project.

The results of the numbering are shown in Fig. 5.26. All except a few devices were numbered. All devices without a main function were not numbered.

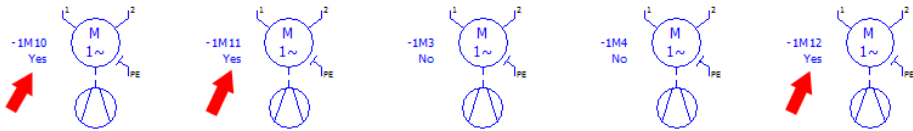


Fig. 5.26 Result of numbering

You can now use **ASSIGN MAIN FUNCTION** to transfer the characteristic to devices without the main function. You first select the devices in the schematic, one after the other, and then select the **ASSIGN MAIN FUNCTION** item in the **PROJECT DATA/DEVICES** menu.

Assign main function

After performing this action (i.e. by clicking the menu item), all devices now have the main function. If you now start the **NUMBER** function again for comparison purposes, the preview dialog will show all devices waiting to receive the new device tag.

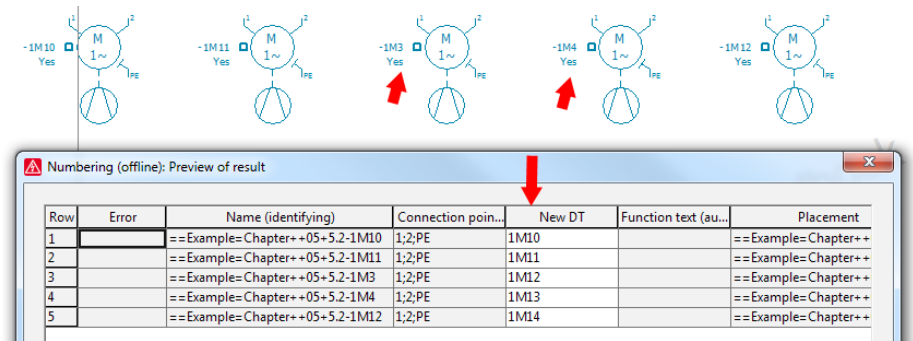


Fig. 5.27 All devices are now numbered.

ASSIGN MAIN FUNCTION can, of course, be assigned to a keyboard shortcut.

5.2.3 Synchronize function texts

This function allows you to synchronize the function texts of the selected devices, including all auxiliary functions and the main function. It makes no difference whether the main function or one of the auxiliary functions is selected.

It must always be the same device, though. Different devices cannot be edited with this function.

All devices should receive the same function text. First you select an element. This can be the main function or any auxiliary function in the schematic.

Then you execute the **SYNCHRONIZE FUNCTION TEXTS** function via the **PROJECT DATA/DEVICES** menu item.



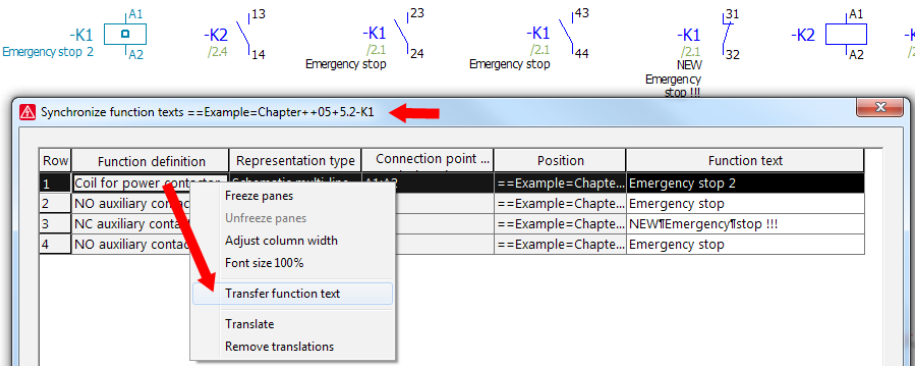


Fig. 5.28 A selected element and the Synchronize function texts dialog

EPLAN opens the **SYNCHRONIZE FUNCTION TEXTS** dialog. In this dialog, only the row containing the function text to be transferred to all other functions is selected. In the example, the function text of row 2 will be transferred to all other elements. Right-click to open the popup menu and select the **TRANSFER FUNCTION TEXT** function. EPLAN now transfers the function text to all other entries in the dialog.

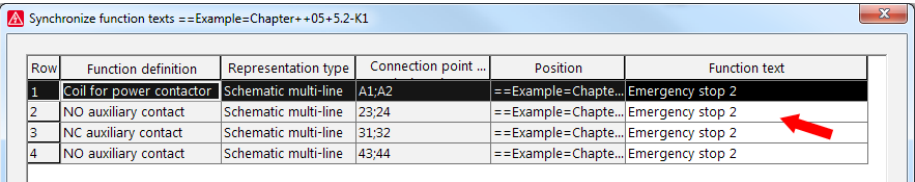


Fig. 5.29 Transferred function text in the dialog

After the transfer, you click the **OK** button. EPLAN exits the dialog and transfers the function text to all elements and writes this text to the **function text <20011>**, **function text (automatic) <20031>**, and **function text (common) <20120>** symbol properties.

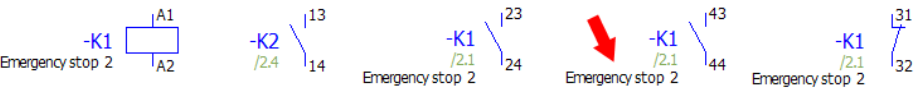


Fig. 5.30 Transferred function text in the schematic

5.2.4 Numbering (Number DT)

EPLAN can number devices according to a definable numbering pattern as soon as they are inserted (according to the personal setting). If the schematic and its devices need to be renumbered in a different format later, a fast and effective un-numbering method is important.

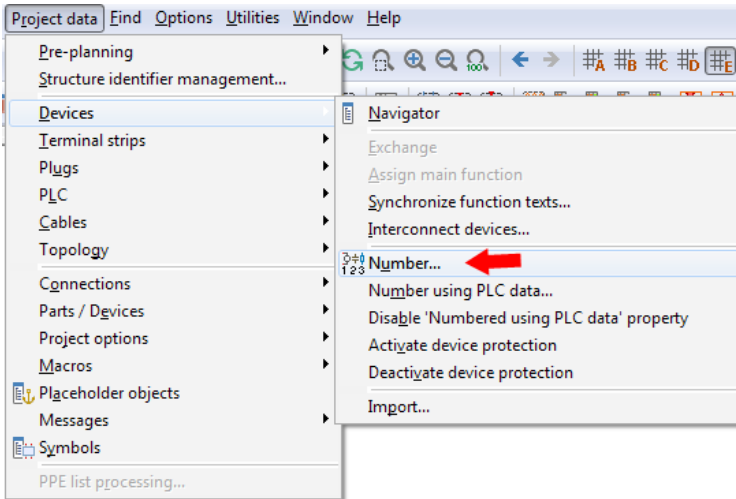


Fig. 5.31 Numbering devices

EPLAN offers the **NUMBER DEVICES** function for this. In contrast to numbering when inserting symbols (**online numbering**), subsequent renumbering is **offline numbering**. Certain filters and schemes can be used for numbering, and predefined schemes can also be applied, exported or imported.

One important thing should be noted about offline numbering. Auxiliary functions cannot be numbered (if there are only auxiliary functions), that is, if there is no main function, then this object will not be included in the numbering. If these items are selected and numbering is then attempted, EPLAN displays an appropriate message.

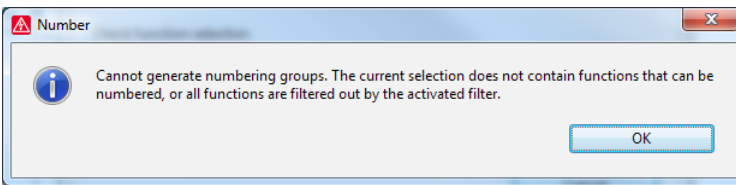


Fig. 5.32 Message

How does the numbering function work? Very simple. You select the devices to be numbered and then start the **NUMBER** function from the **PROJECT DATA/DEVICES** menu.



Number (offline)

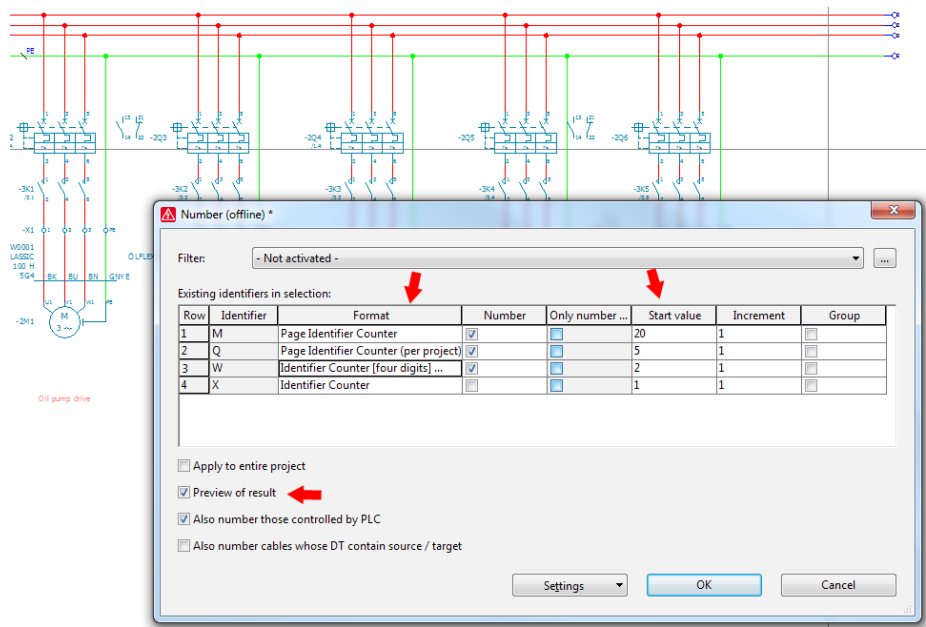


Fig. 5.33 Settings in the Number (offline) dialog

EPLAN opens the **NUMBER (OFFLINE)** dialog and lists the found **identifiers** in a table. In this dialog, you can set your own **numbering format**, start value and increment for each identifier entry. Entire identifiers can be excluded here from numbering so that you are not forced to select only very specific devices when selecting devices.

NOTE: It is recommended to always use the *Preview of result* setting. This enables you to check one more time that everything is numbered (and intervene) as desired before EPLAN writes the new numbering back to the project.

A different numbering format can simply be selected in the *Format* selection list, or the **EXTRAS** button in the lower area of the dialog can be used to create a new numbering scheme.

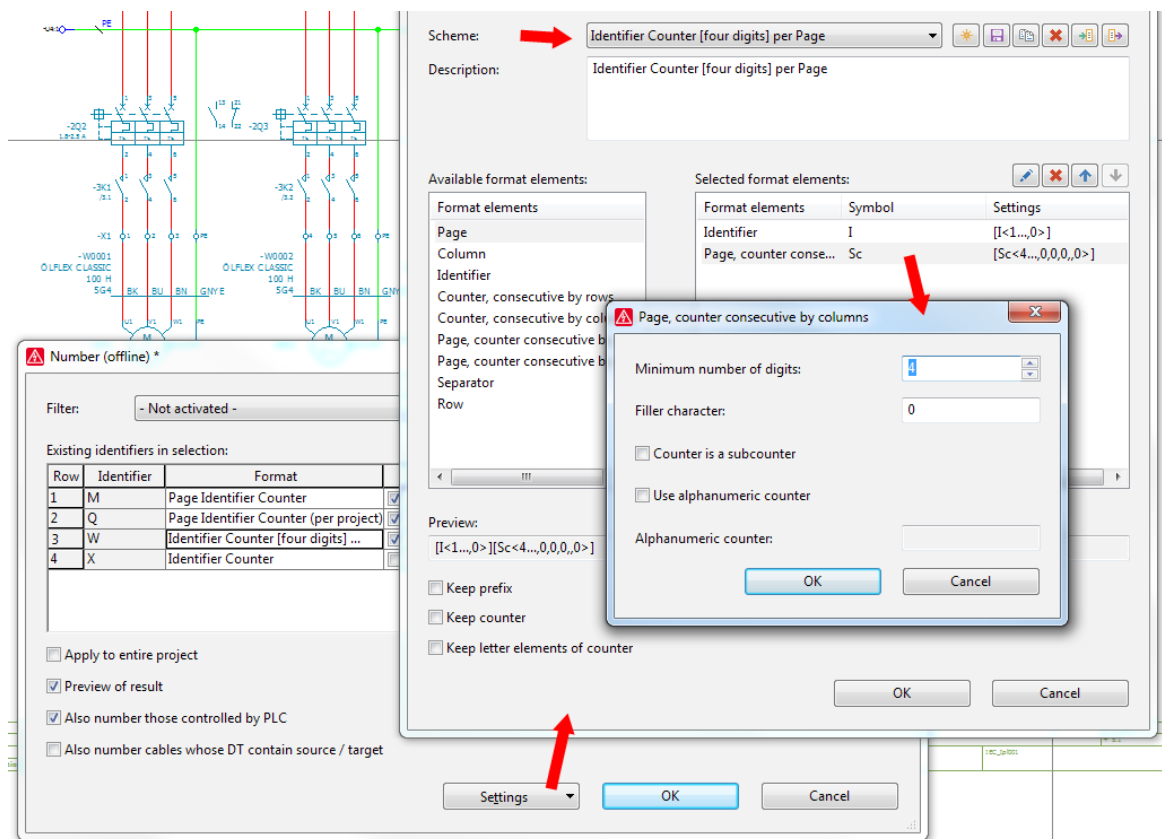


Fig. 5.34 Settings: Numbering (offline) dialog

After you have selected the numbering format, you click OK. EPLAN numbers all devices according to the specified format and displays this numbering in a *preview of result*.

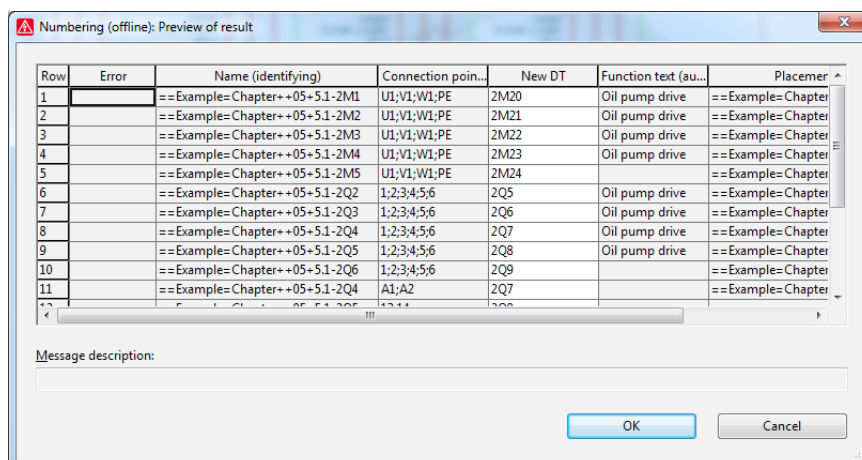


Fig. 5.35 Preview of result dialog

If the **PREVIEW OF RESULT** (if used) is confirmed, then EPLAN writes the modified device tags back to the schematic.

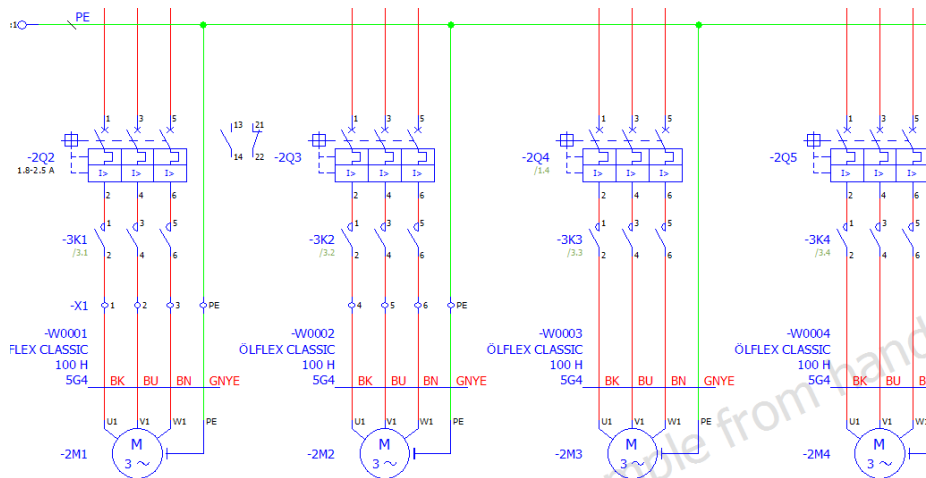


Fig. 5.36 Before numbering

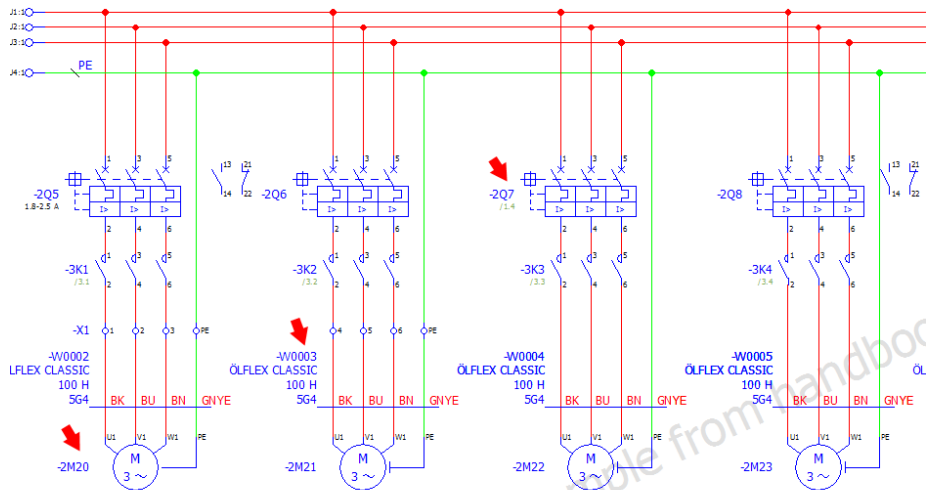


Fig. 5.37 After numbering

In this example, it is obvious that EPLAN only numbers devices that are also a main function. In this example that would be everything except the coil contacts. These were not included in the numbering dialog.

This is why it is important to know that although you can number or re-number devices very quickly by selecting them on a page, these devices must be main functions. If you

want to avoid this, you can perform numbering via the device navigator. Select the devices, right click to open the popup menu and select the **NUMBER DT** menu item.

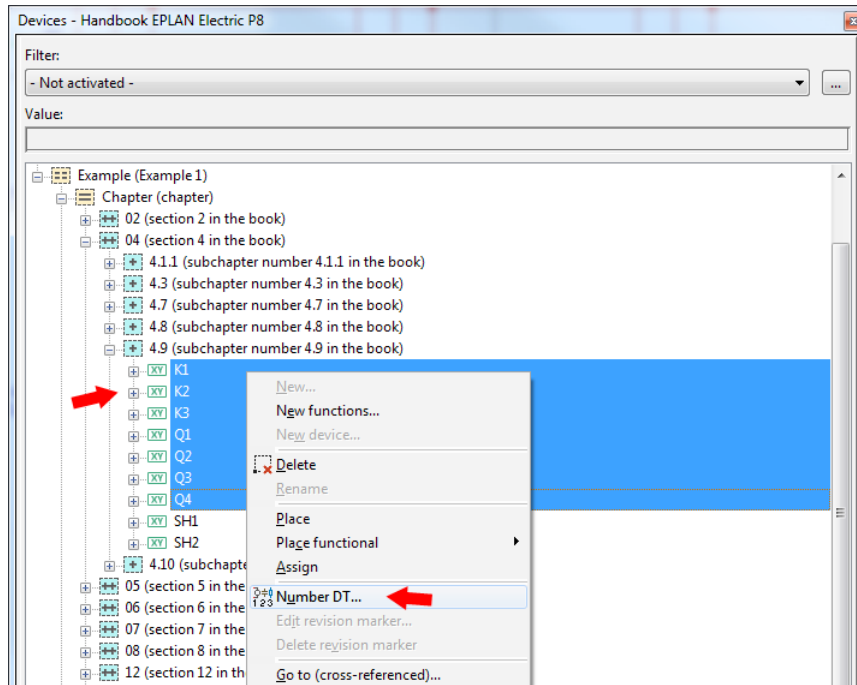


Fig. 5.38 Number DT in the popup menu

Then EPLAN restarts the **NUMBER (OFFLINE)** dialog, and numbering can proceed from here as described above. This time, however, all functions are included because the contacts *and* the coil have been selected (in the device navigator).

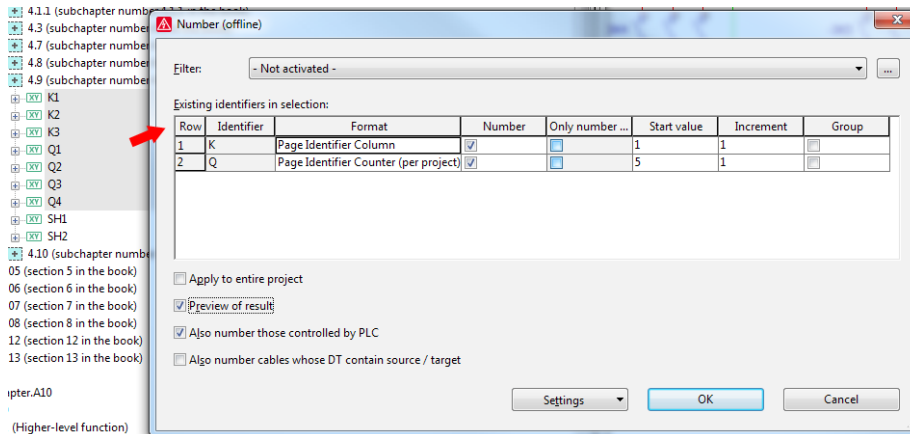


Fig. 5.39 Number (offline) dialog in the device navigator

5.2.5 Device protection

Among other things, device protection makes it possible to protect devices – and, in particular, the part assigned – against modifications. This way, unwanted changes, part or device selections can be prevented.

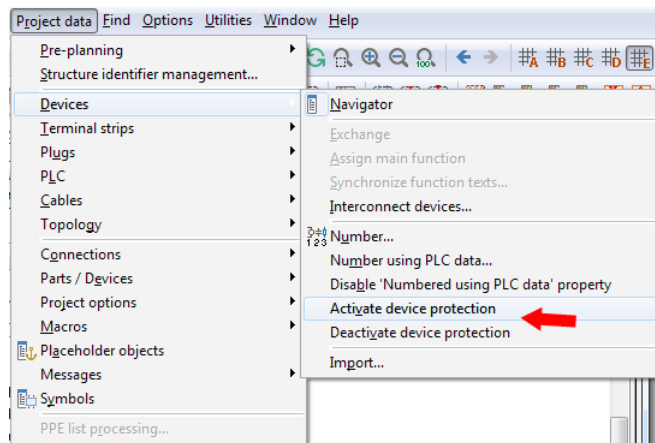


Fig. 5.40 Enabling/disabling device protection

Protected devices can be deleted in a project, but always remain in the project as unplaced devices.

5.2.5.1 Enable

The device protection is enabled simply by selecting the devices to be protected. Then, it is enabled in the **PROJECT DATA/DEVICES/ENABLE DEVICE PROTECTION** menu. EPLAN will then identify such protected devices using a modified icon in the navigators.

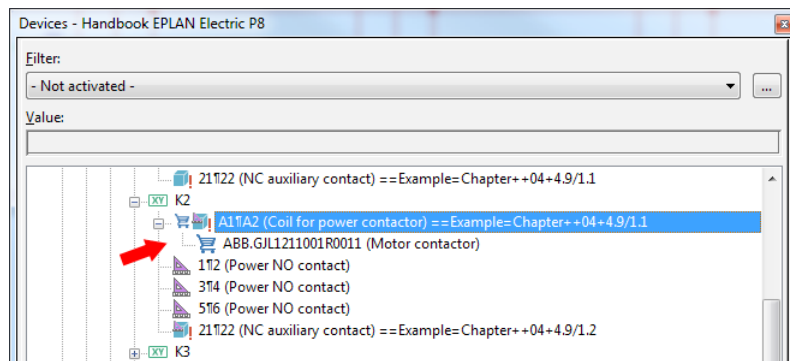


Fig. 5.41 Object without device protection

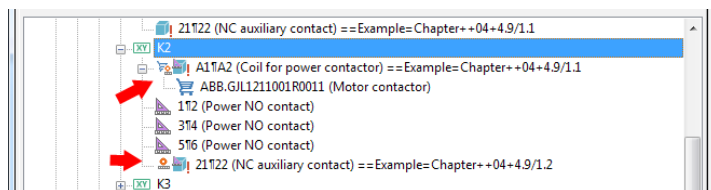


Fig. 5.42 Object with device protection

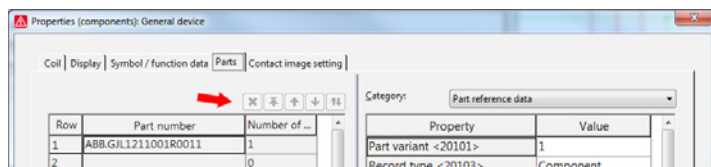


Fig. 5.43 Protected part (Parts tab)

5.2.5.2 Disable

The protection is easily disabled again by selecting the objects and then clicking the **PROJECT DATA/DEVICES/DISABLE DEVICE PROTECTION** menu.

5.2.6 Import (device data)

The **IMPORT** function in the device navigator allows for external data lists to be imported in EPLAN.

C2		ExampleTypeDesignationF22				
	B	C	D	E	F	G
1	Part number	Type designation of part	Order number	Number of units / qu	Remark	Function definition
2	ExamplePartNumberF22	ExampleTypeDesignationF22	ExampleOrderNun	1	NEW	Safety fuse
3	ExamplePartNumberK22	ExampleTypeDesignationK22	ExampleOrderNun	1	NEW	Coil for power contactor
4	ExamplePartNumberQ22	ExampleTypeDesignationQ22	ExampleOrderNun	2	NEW	Fused disconnect, triple
5						
6						

Fig. 5.44 Example of data list (Excel)

An import scheme must be used in EPLAN in order to import the data. EPLAN data fields and external data fields are assigned in this scheme.

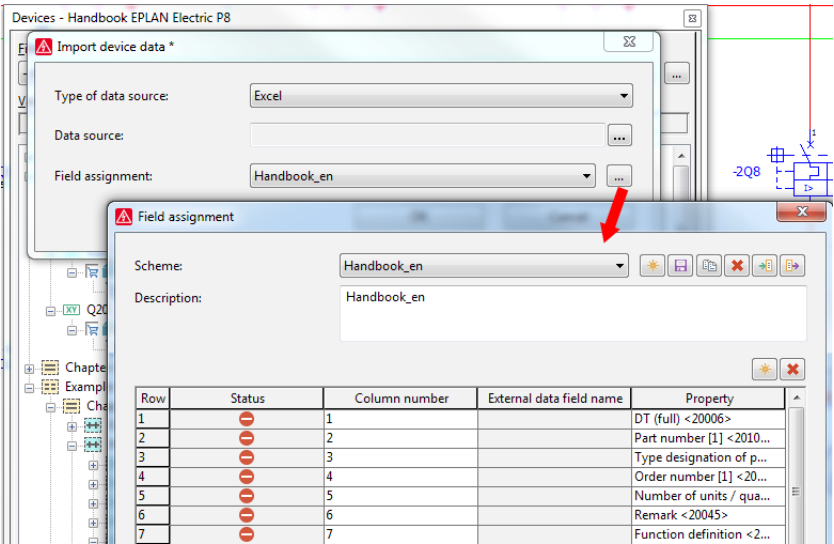


Fig. 5.45 Example of import scheme

A default import scheme can be configured in the settings under **OPTIONS/SETTINGS/PROJECTS/[PROJECT NAME]/DEVICES/IMPORT**.

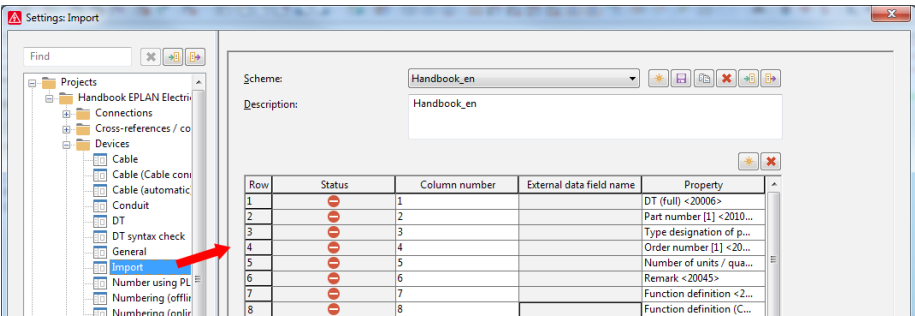


Fig. 5.46 Scheme in project settings

After selecting **IMPORT**, EPLAN opens the **IMPORT DEVICE DATA** dialog. Here, the type of data source, the data source itself and, of course, the import scheme (field assignment) are configured.